AI-POWERED SMART ORCHID CARE SYSTEM FOR VANDA CULTIVATION

INDIVIDUAL THESIS

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Individual component: Disease Detection, Severity Grading and Treatment Recommendation

Group ID: R26-SE-018

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Honours) in Information Technology

Specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Fungal leaf diseases are a frequent cause of loss in Vanda orchid cultivation, yet in Sri Lankan shade houses they are still diagnosed by eye, graded by impression and treated from memory. Published orchid disease classifiers usually report a single high accuracy without describing how training and test images were separated, and stop at a disease name without grading severity, recommending treatment or refusing photographs they cannot interpret. This thesis presents the disease component of an integrated orchid care system. From 667 field photographs of Black Leaf Spot, Phyllosticta Leaf Spot and healthy leaves, a leakage-checked pipeline was built that corrects camera orientation, splits the data before augmentation and holds 67 original photographs out for testing. A MobileNetV2 network pretrained on ImageNet was fine-tuned for diagnosis and a second network grades severity as mild, moderate or severe. A four-stage cascade refuses photographs unlike any training image using a nearest-neighbour feature distance, refers predictions below 0.70 confidence to an expert, grades severity only for an identified disease, and maps disease and grade to a treatment knowledge base that withholds any unverified chemical dose. On the held-out test set the classifier reaches 0.806 accuracy and 0.778 macro F1 against a majority-class baseline of 0.418 and recognises all 24 healthy leaves; MobileNetV2 was selected over EfficientNetB0 by measured comparison. Severity grading is weaker, at 0.465 accuracy against a 0.395 baseline, although 90.7% of grades fall within one step. The input check refused 15 of 17 non-orchid photographs while accepting all 67 test leaves. The component is deployed through a FastAPI service and a mobile application and is consumed at run time by the pollination component. Its contribution is a diagnosis pipeline whose every result is measured on held-out field photographs and which declines to answer when the evidence does not support an answer.

Keywords: orchid disease detection, transfer learning, severity grading, out-of-distribution detection, Vanda

ACKNOWLEDGEMENTS

I am grateful to my supervisor, Ms. Hansi de Silva, and co-supervisor, Ms. Thilini Jayalath, for their guidance throughout this project, and to the evaluation panel, whose questions about evaluation evidence and about the trustworthiness of user-contributed images shaped the final form of this component.

I thank the Vanda growers who allowed me into their shade houses to photograph their plants; without their photographs this component could not have been built. I am also grateful to Mrs. Anoma, Head of Orchid Section at Peradeniya Botanical Garden, for reviewing the disease classes, the treatment recommendations and the contributed photographs.

I also thank my group members, Wickramasinghe D. P., Vithanage P. V. P. M. and Samadinee L. D. N., for the shared backend, application and integration work on which this component depends.

This project received no external funding. The field photographs were taken on my own phone, the models were trained on Google Colab, and the evaluation and testing were run on my own laptop.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CSV	Comma-Separated Values
CWT	Continuous Wavelet Transform
EXIF	Exchangeable Image File Format
F1	Harmonic mean of precision and recall
FRAC	Fungicide Resistance Action Committee
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
kNN	k-Nearest Neighbours
OOD	Out-of-Distribution
QWK	Quadratic Weighted Kappa
RGB	Red, Green and Blue colour channels
SDG	Sustainable Development Goal
UV	Ultraviolet
YOLO	You Only Look Once (object detector)

1 INTRODUCTION

This chapter places the disease component within the group project, states the problem it addresses and the gap in existing work, and sets out the aim, objectives, research questions and scope that the rest of the thesis answers.

1.1 Overall project context

This thesis is one of four individual theses produced by group R26-SE-018, whose common project is an AI-powered smart orchid care system for Vanda cultivation. The system gives a Sri Lankan Vanda grower one mobile application, backed by one FastAPI server and one Firebase database, through which four recurring husbandry decisions are supported by measured recommendations rather than unaided visual judgement. Component 1 (Wickramasinghe D. P.) decides whether a plant is fit to breed from and whether a proposed cross has precedent. Component 2, the subject of this thesis, identifies leaf disease, grades its severity and recommends treatment. Component 3 (Vithanage P. V. P. M.) schedules watering and fertilisation from sensor readings and weather. Component 4 (Samadinee L. D. N.) classifies growth stage and estimates bloom timing.

Vanda orchids sit at the high-value end of Sri Lankan floriculture and are grown mostly by smallholders in shade houses of a few hundred square metres. The knowledge needed to keep them healthy is well documented, but it lives in experienced growers rather than in instruments, and it is applied to each plant from memory. A grower who is new, tired or managing more plants than can be inspected daily will miss things, and the consequences of a missed infection appear only after it has spread.

The group adopted a common evaluation discipline that this component follows throughout: every reported figure is stated beside the majority-class baseline of the same held-out split, every split reflects the conditions of deployment, results that fall short are reported rather than omitted, and the system must refuse inputs it cannot interpret rather than answer them confidently. The common integrated solution report

describes how the four components were joined; this thesis describes Component 2 in full.

1.2 Individual component context

Component 2 answers the second of the project's four questions: is this plant diseased, with which disease, and how badly? A grower photographs a leaf with the phone camera or selects a photograph from the gallery, and the component returns a disease verdict with a confidence, a severity grade when a disease is identified, and a treatment recommendation matched to the disease and the grade. The same verdict is consumed at run time by Component 1, because a diseased plant should not be selected as a breeding parent. Figure 1.1 places the component in the wider system.

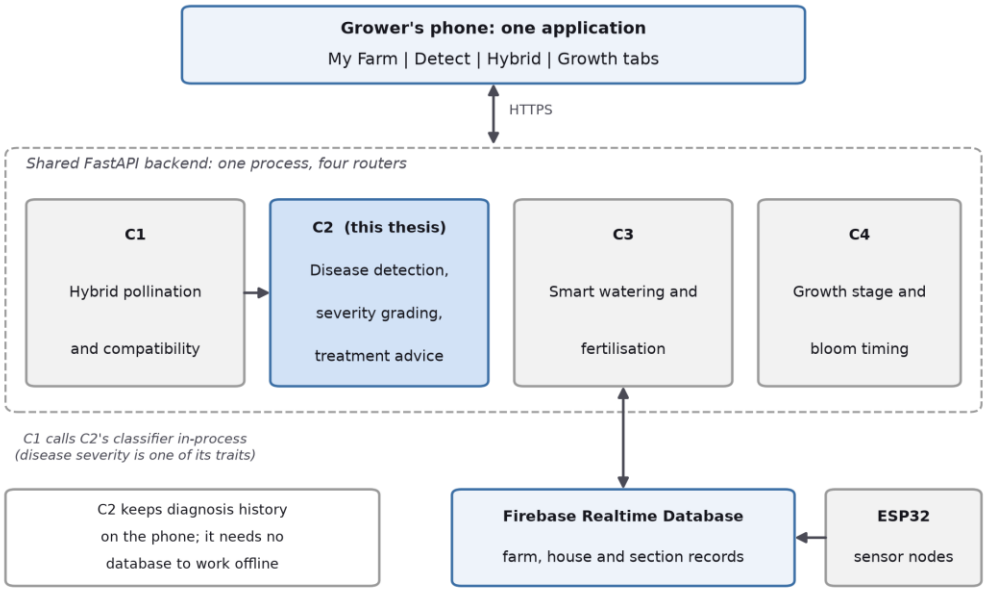


Figure 1.1: Component 2 within the integrated solution. Component 1 calls Component 2's classifier at run time instead of training a disease model of its own

Leaf diseases matter for Vanda because the plants are grown densely under conditions that favour fungi. Phyllosticta leaf spot spreads in splashing water, so overhead irrigation is its main vector, and both fungal leaf spots studied here are favoured by prolonged leaf wetness and poor air movement. A lesion that is identified early can be removed and the plant isolated before a bench is affected; a lesion that is misidentified may be treated with the wrong product or not at all.

The project originally planned five disease classes: Phyllosticta leaf spot, Black leaf spot, Anthracnose, Phytophthora black rot and Sooty mould. The scope was narrowed to the two diseases for which enough real field photographs could be collected, Black Leaf Spot and Phyllosticta Leaf Spot, together with a healthy class. Fusarium wilt and Phytophthora black rot had only 13 and 20 usable photographs respectively, far too few to train on, and the remaining classes were deferred [state the reason Anthracnose and Sooty mould were deferred]. Conditions outside the three trained classes are handled by referral rather than by a trained catch-all class, and photographs of new diseases can be contributed through the application for future training, as described in Chapter 4.

1.3 Problem statement

A Vanda grower who finds a marked leaf needs three answers at once: which disease it is, how far it has progressed, and what to do. Each is hard to obtain in the shade house. The two common fungal leaf spots look alike to a non-specialist, and Chapter 6 shows that they are also the pair the trained model confuses most. Severity is judged by impression, so two growers grade the same leaf differently. Treatment advice circulates without sources, and a chemical rate repeated from memory can be wrong for the product sold locally. Expert help exists in extension services and research institutes, but it cannot be consulted at the moment a decision is taken.

A mobile classifier appears to solve this, but it introduces problems of its own. A network trained on a few hundred photographs must always choose one of its trained classes, so it will diagnose any image it is shown. During development, a photograph of a meal was classified as a healthy orchid leaf with 99.9% confidence, grass scored healthy at 99.3%, and a tree was diagnosed with Phyllosticta at 99.6%. A system that answers every question with high confidence is more dangerous in a shade house than one that sometimes says it does not know.

The problem addressed by this component is therefore stated as follows: how can a phone photograph of a Vanda leaf be turned into a trustworthy disease verdict, severity grade and treatment recommendation from only a few hundred labelled field

photographs, while refusing photographs the system cannot interpret and never presenting advice that has not been verified?

1.4 Research gap

The literature reviewed in Chapter 2 contains capable orchid disease work, but five gaps recur. First, reported accuracy is rarely accompanied by a description of how training and test images were separated or by a baseline. A two-class orchid leaf classifier reports 98% accuracy [1], a viral disease system reports 91.8% [2], and an integrated orchid system reports detection rates of about 90% [3]; none of these figures can be interpreted without knowing the class balance and whether variants of the same leaf appeared on both sides of the split. General plant disease work shows why this matters: a model reaching 99.35% on held-out images of one dataset fell to just above 31% on images from other sources [4].

Second, the reviewed orchid systems name a disease but do not grade its severity, although severity is what decides between pruning a leaf and treating a whole bench. Third, treatment advice is either absent or presented without a verifiable source; the integrated system above lists fungicides per disease [3] but does not tie dose to severity or to a registered product label. Fourth, none of the reviewed image-based systems describes refusing an image it was never trained to interpret, although softmax confidence is known to be an unreliable signal for unfamiliar inputs [5]. Fifth, the class list is fixed at training time and no reviewed system offers a safe, verified route by which a new disease can be added.

The work is also concentrated on other genera and countries: Phalaenopsis in Taiwan [2], [6], [7], commercial orchids in Thailand [1] and black orchids in Indonesia [8]. No reviewed study used field photographs of Vanda grown in Sri Lankan shade houses. Section 2.5 synthesises these gaps and maps each to a design response.

1.5 Aim and objectives

The aim of this component is to design, build and evaluate a mobile disease-diagnosis component for Vanda orchids that identifies the disease from a leaf photograph, grades its severity, recommends a treatment from a verifiable knowledge base, and declines to answer when the evidence does not support an answer.

Six objectives follow from that aim.

- O1: Assemble a labelled, leakage-free dataset of Vanda field photographs with disease and severity labels and a reproducible preparation pipeline.
- O2: Develop a disease classifier, select its architecture by measured comparison, and evaluate it on held-out photographs against a majority-class baseline.
- O3: Develop a severity grader for identified diseases and evaluate it with measures suited to an ordered scale.
- O4: Build an input check that refuses photographs unlike the training data, and a confidence rule that refers uncertain predictions to an expert.
- O5: Build a treatment knowledge base keyed by disease and severity that never displays an unverified chemical dose.
- O6: Deploy the component through the shared backend and mobile application, integrate it with Component 1, and provide a verified route by which photographs of new diseases can be collected for retraining.

Table 1.1 maps each objective to the sections that address it and to the evidence that settles it.

Table 1.1: Objectives, the sections that address them and the evidence that settles each

Objective	Addressed in	Evidence
O1 Leakage-free dataset	Sections 3.2 and 3.3.1; Appendix B	667 field photographs; split before augmentation; three-way leakage check passed
O2 Disease classifier	Sections 3.3.2, 3.3.3 and 6.1.1 to 6.1.3	0.806 accuracy and 0.778 macro F1 against a 0.418 baseline; backbone chosen by measured comparison
O3 Severity grader	Sections 3.3.5 and 6.1.5	0.465 exact accuracy against a 0.395 baseline; 90.7% of grades within one step
O4 Input check and confidence rule	Sections 3.3.4, 3.3.6, 6.1.4 and 6.1.6; Appendix D	15 of 17 non-orchid photographs refused and every test photograph accepted; threshold 0.70 chosen on validation
O5 Treatment knowledge base	Section 4.4.3; Appendix C	no unverified dose displayed in any run (Table 6.6)
O6 Deployment, integration and verified contributions	Sections 4.4.6, 5.3, 5.5, 6.2 and 6.4	service and application tested end to end; Component 1 consumes the cascade; expert review in progress

1.6 Research questions or hypotheses

The objectives are examined through five research questions.

- RQ1: How accurately can a transfer-learned convolutional neural network distinguish Black Leaf Spot, Phyllosticta Leaf Spot and healthy Vanda leaves in field photographs that contributed nothing to its training, relative to a majority-class baseline?
- RQ2: Is transfer learning necessary at this data scale, and which backbone gives the best balance between accuracy and deployability?
- RQ3: How well can severity be graded from the same photographs, and which errors dominate?
- RQ4: Can a distance in the network's feature space separate orchid leaf photographs from unrelated images where softmax confidence cannot, and at what cost in refused genuine photographs?
- RQ5: Which confidence threshold balances automatic answers against referrals to an expert?

1.7 Scope limitations and assumptions

The component is scoped to Vanda orchids, to leaf diseases, and to the three trained classes. Diseases of stems, roots and flowers, pests and viral infections are outside the classifier's scope; photographs of them may be contributed for future training but are not diagnosed. Laboratory confirmation of a pathogen is outside scope, as is any chemical dose that has not been verified against the product label and the registered list in Sri Lanka.

Three assumptions are load-bearing. The grower photographs a single leaf in daylight at roughly arm's length, which matches the conditions of the training photographs. The grower can act on advice such as isolating a plant or removing a leaf. And any chemical treatment is applied according to the product label, which is the legal instruction.

The principal limitations, examined in Section 7.5, are the size of the dataset (667 originals, 67 of them held out for testing), the fact that severity labels were assigned

by one person, the small set of non-orchid photographs used to calibrate the input check, and a split that was stratified by disease before severity labels existed.

1.8 Individual contribution

The work described in this thesis is the author's individual contribution to the group project. It comprises the collection and labelling of the photographs [confirm that you took the photographs and assigned every label]; the data preparation tools for orientation checking, renaming, augmentation, label templates, severity labelling and leakage checking; the training and evaluation scripts for both models and the architecture comparison; the inference cascade and the feature-distance input check; the treatment knowledge base and the advisor that enforces its safety rule; the backend router and service; and the four mobile screens and two services that make up the disease section of the application.

The integration through which Component 1 consumes this component's classifier was implemented by the Component 1 owner against the interface this component exposes. The shared backend, application shell and deployment were developed jointly. Use of generative AI tools during development and writing is disclosed in Appendix G.

1.9 Thesis structure

Chapter 2 reviews the domain, the methods used and the related orchid disease work, and synthesises the gaps into a conceptual framework. Chapter 3 describes the research design, data, procedures, metrics and ethical considerations. Chapter 4 sets out the requirements and the design of the component within the complete solution. Chapter 5 describes implementation, integration with the other components, verification and deployment. Chapter 6 reports the measured results, and Chapter 7 interprets them against the literature and states the limitations. Chapter 8 considers commercialisation and sustainability, and Chapter 9 concludes and sets out future work.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the plant pathology and machine learning foundations the component rests on, the orchid-specific disease research it builds on, and the existing systems it can be compared with.

2.1 Review method and scope

The review was structured rather than systematic. Sources were located through IEEE Xplore under the institution's subscription, Frontiers in Plant Science and Google Scholar, using combinations of the terms orchid disease detection, orchid leaf disease classification, plant disease deep learning, transfer learning, disease severity assessment and out-of-distribution detection. Applied work was restricted to 2016 to 2025 so that it reflects current deep-learning practice; foundational method papers were included regardless of date.

Studies were included if they detected orchid disease by any modality, if they were influential deep-learning studies of plant disease that bear on field deployment, or if they define a method used in this component. Eight orchid-specific studies were retained and are summarised in Table 2.1. The review is limited by its reliance on English-language, peer-reviewed sources and by the small size of the orchid disease literature.

2.2 Domain and theoretical foundations

2.2.1 Plant pathology

The two diseases studied are fungal leaf spots. Phyllosticta leaf spot is caused by Phyllosticta species, whose sexual stage is classified as Guignardia; it is favoured by warm, persistently humid conditions with water on the leaf, and its spores are spread by splashing water. Black leaf spot is treated here as a fungal leaf-spotting complex favoured by prolonged leaf wetness, crowding and poor air movement. The distinction between fungi and oomycetes matters for treatment: Phytophthora, which causes black rot in orchids, is an oomycete rather than a true fungus and responds to different chemistry, such as metalaxyl-class products, than the protectant fungicides used against leaf spots [9]. Although Phytophthora is outside the classifier's scope,

the treatment knowledge base retains this distinction so that it can never recommend a fungicide for an oomycete.

2.2.2 Severity assessment

Disease severity is conventionally expressed as the proportion of plant tissue showing symptoms. Visual estimates of that proportion are fast but vary between raters, and their reliability improves when raters work to defined categories or reference diagrams rather than free estimates [10]. This component adopts three categories defined by affected leaf area: mild below 10%, moderate from 10% to 40%, and severe above 40%. Because the categories are cut-points on a continuous quantity, a leaf near a boundary can reasonably fall either side of it, which is why the severity grader is evaluated with a within-one-grade measure as well as exact accuracy.

2.2.3 Transfer learning

Convolutional neural networks trained on ImageNet, a database of over a million labelled photographs [11], learn early filters that respond to edges, colours and textures and later filters that respond to object-specific patterns. Features in the early layers transfer well to new tasks while the later layers are more task-specific [12], which is why a small dataset is best served by keeping the early layers and adapting only the last ones. MobileNetV2 builds its network from inverted residual blocks with depthwise separable convolutions, which reduces computation and model size for mobile deployment [13]. EfficientNet scales depth, width and resolution together and achieves strong accuracy per parameter [14]. Both are evaluated in Chapter 6.

2.2.4 Evaluation under imbalance and leakage

Data leakage occurs when information about the test set is available during training, and it inflates reported performance [15]. In image classification the commonest form is placing augmented copies or near-duplicate photographs of the same leaf on both sides of a split. Under class imbalance, overall accuracy can also mislead, because a model that always predicts the largest class scores that class's share; macro-averaged F1, which weights every class equally, and the majority-class baseline are therefore reported alongside accuracy. For an ordered scale such as

severity, weighted kappa penalises a two-grade error more than a one-grade error [16].

2.2.5 Out-of-distribution detection

A softmax classifier distributes probability only across the classes it was trained on. Maximum softmax probability is a common baseline for detecting unfamiliar inputs [5], but it measures how clearly an input fits one trained class better than the others, not whether the input resembles the training data at all. Distance-based methods instead compare an input's internal feature vector with those of the training data; a non-parametric nearest-neighbour distance in the penultimate feature space has been shown to detect out-of-distribution inputs without distributional assumptions [17]. This component uses that approach.

2.3 Related research

2.3.1 Image-based orchid disease classification

Santisarn and Titijaroonroj collected leaf images on a Thai orchid farm, divided into normal leaves and leaves with yellow leaf spot, compared several pretrained networks including DenseNet201, GoogLeNet and MobileNetV2, and reported the highest accuracy, 98%, for MobileNetV2; the model was deployed behind a LINE chatbot that notifies and tracks discovered disease [1]. Tsai et al. identified viral diseases of orchids from leaf images through colour-space transformation, gamma correction, U-Net leaf segmentation, texture features and a two-stage combination of a random forest and an Inception network, reporting 0.9707 accuracy for leaf segmentation and 0.9180 for disease identification, and outperforming naked-eye identification for the easily confused Cymbidium mosaic and Odontoglossum ringspot viruses [2]. Ismayanti et al. identified four pests of the Indonesian black orchid from leaf texture features (contrast, correlation, energy and homogeneity) with a naive Bayes classifier [8].

2.3.2 Non-RGB and environmental approaches

Wu et al. detected Fusarium wilt in Phalaenopsis from per-pixel hyperspectral data in the visible-near-infrared and short-wave infrared ranges with a lightweight network, EE-CSPNet, reporting 98.12% and 97.64% accuracy and a 97% reduction in

parameters against a Darknet-53 baseline [7]. A Taiwanese group has approached the problem without images: Chen et al. transformed greenhouse temperature and humidity series with a continuous wavelet transform and fed them to a 3-D ConvLSTM to detect rare disease events [6]; Lin et al. designed a spore germination sensor with dynamic calibration for large greenhouses [18] and combined it with the wavelet pipeline in OrchidTalk-v2, reporting precision above 93% and recall above 92.75% [19]. These systems estimate the risk of an outbreak in a house rather than diagnosing a particular leaf, and are therefore complementary to image-based diagnosis.

2.3.3 Deep learning for plant disease in general

Mohanty et al. trained networks on 54,306 images of 14 crops and 26 diseases and reached 99.35% accuracy on held-out images of the same collection, but accuracy fell to just above 31% on images gathered from other sources [4]. Ferentinos trained on 87,848 images covering 58 plant and disease classes and reported a best success rate of 99.53%, while noting that models trained on laboratory images performed considerably worse on field images [20]. Barbedo showed that the size and variety of the dataset strongly limit how well deep learning and transfer learning generalise for plant disease classification [21]. Together these results explain the design choices of this component: field photographs rather than laboratory images, transfer learning because the dataset is small, and a held-out test set of original photographs because accuracy on augmented or same-source images overstates performance.

2.4 Existing systems products and techniques

The closest existing system is OrchiZen, an integrated smart farming system for Dendrobium, Vanda and Phalaenopsis developed at the same institution [3]. It combines mist irrigation, species recommendation, ultraviolet disease treatment with light optimisation, and fertiliser recommendation by growth stage. Two YOLOv5 models detect fungal diseases (Phytophthora, Anthracnose and Fusarium) and classify the species, with reported detection rates of 91%, 87% and 90% for the three genera; detected infections are treated with UV-C light at 30 J/cm², which the authors report reduced fungal symptoms by 72% within two cycles, and a table maps each

disease to a fungicide. The paper does not describe a held-out split or a baseline, does not grade severity, and does not describe refusing unrelated images.

The chatbot system described in Section 2.3 is an example of a deployed product: the model runs on a server behind a REST interface and growers interact through a messaging application they already use [1]. General-purpose plant identification applications are free and widely installed, but they identify species and offer generic care text rather than assessing a disease against a decision. Agricultural extension services provide exactly the expertise required but cannot be consulted at the moment a grower is standing at the bench.

Table 2.1: Comparison of the reviewed orchid disease studies with Component 2

Study	Task	Modality and method	Reported result	Severity graded	Refuses unfamiliar input
Santisarn and Titijaroonroj [1]	orchid leaf, 2 classes	RGB images; pretrained CNNs, MobileNetV2 best	98% accuracy	not reported	not described
Tsai et al. [2]	orchid viral diseases	RGB; U-Net, texture features, random forest and Inception	0.918 identification accuracy	not reported	not described
Ismayanti et al. [8]	black orchid pests, 4 classes	RGB leaf texture; naive Bayes	not comparable	not reported	not described
Wu et al. [7]	Phalaenopsis Fusarium wilt	hyperspectral VNIR and SWIR; EE-CSPNet	98.12% and 97.64% accuracy	not reported	not described
Chen et al. [6]	greenhouse outbreak risk	temperature and humidity; CWT and 3-D ConvLSTM	rare-event detection	not applicable	not applicable
Lin et al. [19]	greenhouse outbreak risk	spore sensor; CWT and 3-D ConvLSTM	precision above 93%, recall above 92.75%	not applicable	not applicable
OrchiZen [3]	fungal disease, 3 genera	RGB; two YOLOv5 models; UV-C treatment	detection rates 87% to 91%	not reported	not described
This component	Vanda leaf, 3 classes	RGB field photographs; MobileNetV2 cascade	0.806 accuracy, 0.778 macro F1 (baseline 0.418)	yes	yes

Table 2.1 shows that the reviewed orchid work reports high single figures under conditions that differ from this component's, and that no reviewed system combines diagnosis, severity, sourced treatment and refusal of unfamiliar inputs.

2.5 Gap synthesis and conceptual framework

The review identifies five gaps, each of which this component answers with a specific design response (Figure 2.1).

- G1, evaluation that may leak: answered by splitting the original photographs before augmentation, checking the split three ways, and reporting every figure on held-out originals beside a majority-class baseline (O1, O2).
- G2, confident answers on unfamiliar images: answered by a feature-distance input check and a confidence threshold chosen on the validation split (O4).
- G3, no severity: answered by a separate severity grader evaluated with exact and within-one-grade accuracy (O3).
- G4, unsourced treatment advice: answered by a knowledge base keyed by disease and grade whose unverified doses are withheld (O5).
- G5, a fixed class list: answered by a contribution queue in which photographs of new diseases are verified by an expert before they can be used for retraining (O6).

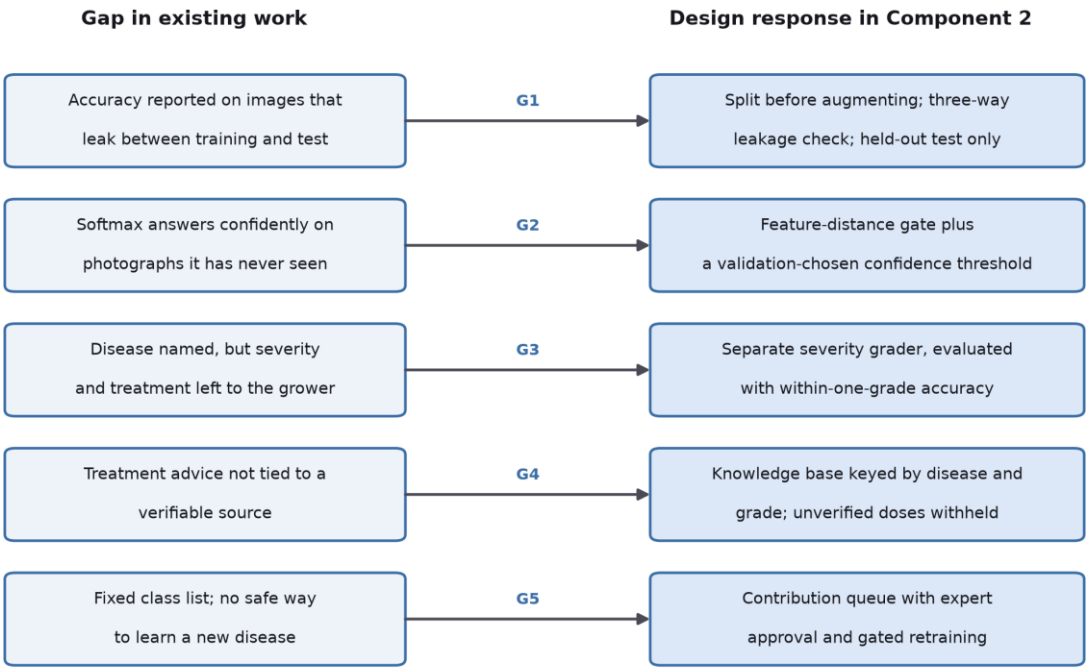


Figure 2.1: Conceptual framework. Each gap found in the reviewed work is answered by a specific design response in Component 2

3 RESEARCH METHODOLOGY

This chapter describes how the component was researched and evaluated: the research design, the data, the procedures for preparing data and training models, the measures used, and the ethical and reproducibility provisions.

3.1 Research design

The component follows a design-science research approach: an artefact is built to address a practical problem, evaluated quantitatively against explicit criteria, and revised in response to what the evaluation reveals (Figure 3.1). Evaluation is quantitative and uses held-out data; qualitative validation by an orchid researcher and by growers complements it (Section 6.3).

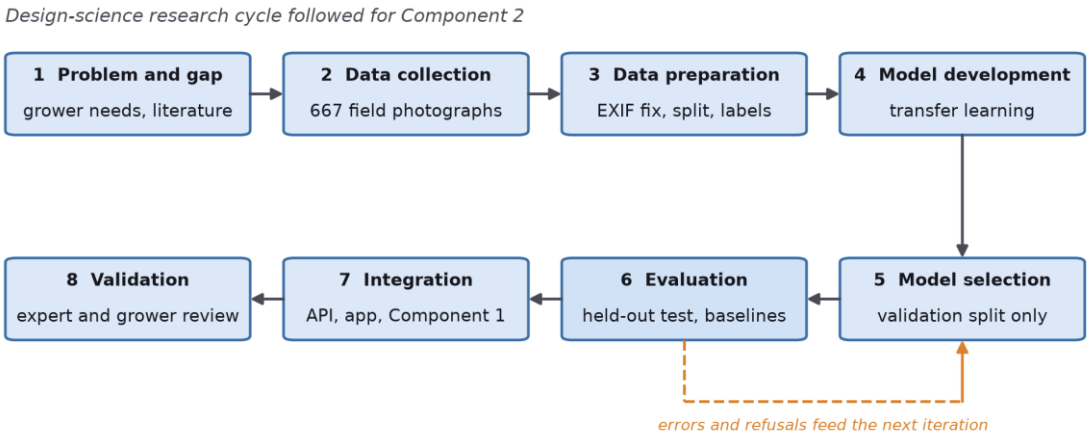


Figure 3.1: Research design. The build and evaluate steps iterate, and evaluation findings drive the next model iteration

Two methodological rules were fixed before any result was collected. All choices of architecture, threshold and checkpoint are made on the validation split and never on the test split, because choosing on the test split would tune the reported result. And the test split is used once per model, after every choice has been made. The iteration shown in Figure 3.1 was not hypothetical: evaluation exposed a 10.4 percentage point accuracy loss caused by a preprocessing mismatch and the confident misclassification of unrelated photographs, and each led to a design change described in Chapter 5.

3.2 Data participants or study materials

The primary material is a set of 667 field photographs of Vanda leaves taken with a phone camera in growers' shade houses [state the collection sites, the period of collection and the phone model]. Photographs were taken with the growers' permission and show plants, not people. Each photograph was assigned to one of three classes, and every diseased photograph was graded for severity (Table 3.1).

Table 3.1: Composition of the dataset and of each split. Training files after augmentation are 54 per original

Class	Originals	Train	Validation	Test	Train after augmentation	Severity grades (mild / moderate / severe)
Black Leaf Spot	152	122	15	15	6,588	52 / 59 / 41
Phyllosticta Leaf Spot	275	219	28	28	11,826	57 / 81 / 137
Healthy	240	192	24	24	10,368	none
Total	667	533	67	67	28,782	109 / 140 / 178

Disease labels were assigned by [state who assigned or confirmed the disease labels, and how]. Severity labels were assigned by the author to all 427 diseased photographs by estimating the proportion of leaf area showing lesions, chlorosis or necrosis and placing it in one of the three bands defined in Section 2.2 [confirm how the proportion was estimated]. The 240 healthy photographs carry no grade. The resulting grade mix among diseased originals is 109 mild, 140 moderate and 178 severe. The grading protocol and the grade counts by disease are given in Appendix B.

Two further sets were used only to calibrate and verify the input check. Seventeen non-orchid photographs (dogs, grass, hands, tables, trees, messaging-application images and a screenshot) represent the kinds of mistaken upload a grower might make. Eleven orchid disease photographs taken from the web, by other people on other equipment, represent genuine orchids that the system should accept although they differ from the training conditions. Neither set was used to train any model.

The human participants in the study are growers who allowed their plants to be photographed and an orchid researcher who reviews the disease classes, treatment recommendations and contributed photographs [state the number of participants, their roles and the dates of consultation].

3.3 Experimental or study procedure

3.3.1 Data preparation

The procedure is summarised in Figure 3.2. A survey of the raw photographs found that 330 of the 667 (49%) were stored sideways, with an orientation tag that photo viewers obey but that a network reading raw pixels ignores; none was corrupt and none was a byte-identical duplicate. Each photograph was therefore rotated so that its stored pixels are upright, converted to RGB, resized so that its longer side is at most 1,024 pixels, and saved under a traceable name, reducing the collection from 1.5 GB to 135 MB. A rename map records the original camera filename behind every processed file.

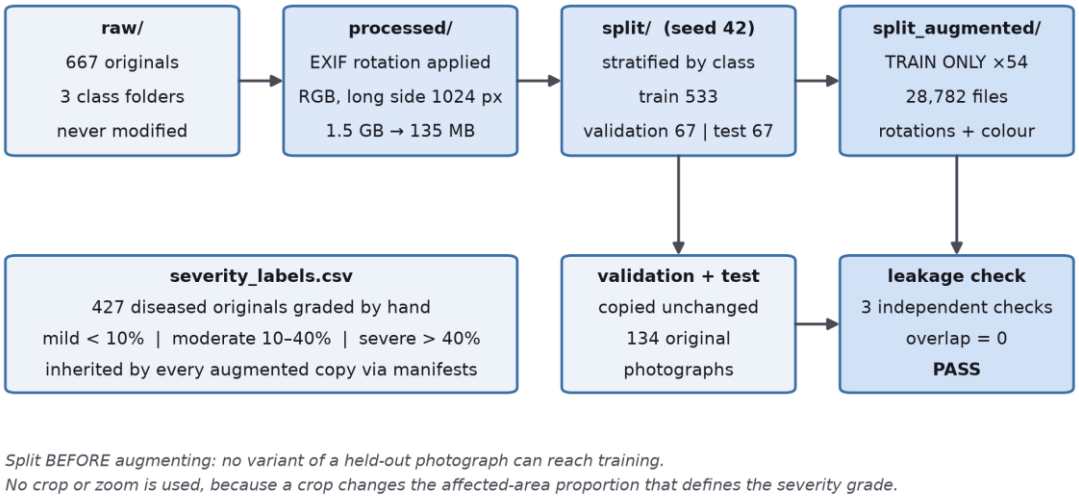


Figure 3.2: Data preparation pipeline. The original photographs are split before augmentation, so no variant of a held-out photograph can reach training

The processed photographs were split 80:10:10 into training, validation and test sets, stratified by class and seeded (seed 42) so that the split is reproducible. Only the training set was then augmented. Each training original produced 54 files: the original, five rotations (45, 90, 180, 225 and 270 degrees), and eight colour adjustments (brightness, exposure, contrast and saturation, each raised and lowered) applied to each of those six images. Rotations by 45 and 225 degrees leave black corners, which a network can learn as a shortcut, so the largest clean rectangle was cropped from the centre instead. Exposure was applied as a gain in linear light so that

it is not a duplicate of brightness. No crop or zoom transform was used, because a crop changes the proportion of diseased tissue and would silently invalidate the severity label inherited by every augmented copy. The validation and test sets were copied unchanged, so they contain only original photographs.

A leakage check then compared the training files with the held-out sets in three independent ways: the source image recorded for every augmented file, the filenames actually present on disk, and the split manifest itself. All three found no overlap; the check was repeated on 10 October 2026 with the same result.

3.3.2 Disease classifier

The classifier is MobileNetV2 with ImageNet weights, a global average pooling layer, dropout of 0.3 and a three-way softmax output. The input is a 224 by 224 RGB image with values from 0 to 255; scaling to the range the network expects is performed by a layer inside the model, so that the deployed file cannot be given doubly scaled input. Training used two stages (Table 3.2). The first trains only the new output layers on the frozen network at a learning rate of 0.001, because large gradients from a randomly initialised head would damage the pretrained filters. The second unfreezes the top 30 layers at a learning rate of 0.00001 so that the late layers adapt to orchid lesion texture, while batch-normalisation layers stay frozen because updating their statistics on small batches destabilises training. Class weights computed by the balanced formula offset the 1.8-to-1 imbalance between the two diseases.

Table 3.2: Training configuration of the disease classifier

Setting	Stage 1 (frozen network)	Stage 2 (fine-tuning)
Trainable layers	new head only	top 30 layers of MobileNetV2 and head
Learning rate	0.001 (Adam)	0.00001 (Adam)
Maximum epochs	25	12
Epochs actually run	16	5
Batch size	32	32
Regularisation	dropout 0.3	dropout 0.3; batch normalisation frozen
Class weights	Black Leaf Spot 1.456, healthy 0.925, Phyllosticta 0.811	same
Callbacks	early stopping and checkpoint on validation loss (patience 5); learning rate halved on plateau	same
Platform	Google Colab, NVIDIA T4 GPU,	same

Setting	Stage 1 (frozen network)	Stage 2 (fine-tuning)
	TensorFlow 2.20	

3.3.3 Architecture comparison

To answer RQ2, MobileNetV2, EfficientNetB0 and a small network trained from scratch were trained under identical conditions (five epochs, frozen pretrained layers, learning rate 0.001, batch size 32, dropout 0.3, seed 42), each with its own official input preprocessing, and were scored on the validation split. The network trained from scratch is a control that shows how much of the performance is due to ImageNet pretraining.

3.3.4 Confidence threshold

To answer RQ5, the share of photographs answered automatically and the accuracy on those answers were computed for thresholds from 0.40 to 0.90 on the validation split, and the operating threshold was chosen from that sweep before the test split was examined.

3.3.5 Severity grader

The severity grader uses the same architecture and recipe but is trained only on the 427 diseased originals, because a healthy plant has no grade. Its labels come from the severity table and are inherited by the augmented copies through the augmentation manifests, which yields 18,414 training files. Balanced class weights (mild 1.536, moderate 1.015, severe 0.733) were applied, and the grader was trained with the pretrained layers frozen, for at most 12 epochs.

3.3.6 Input check

The input check takes the 1,280-value feature vector produced by the classifier's pooling layer, before the softmax forces a choice among the three classes. The vectors of the 533 training originals are stored as a reference bank. For a new photograph the check computes one minus the mean cosine similarity to its five most similar reference photographs, and refuses the photograph when that distance exceeds a threshold. Five neighbours rather than one are used so that a single unusual training photograph cannot admit an unrelated image. The threshold was chosen from the measured separation between orchid and non-orchid photographs, as

reported in Section 6.1. Because nothing is retrained, the check does not change any classification result.

3.3.7 Improvement experiments

Following the panel's feedback at the second progress presentation, a series of controlled experiments is being run to document how each design choice affects accuracy. Each experiment changes one factor and is scored on the validation split: number of epochs; learning rate; augmentation on or off; class weights on or off; frozen against fine-tuned networks at two depths; checkpoint selection by validation macro F1 instead of validation loss; averaging predictions over flipped and rotated copies at prediction time; the backbone comparison repeated with fine-tuning; and, for severity, fine-tuning and a split stratified by both disease and severity. The final models will be evaluated once on the test split and with five-fold cross-validation.

3.4 Measures and evaluation metrics

Table 3.3 defines the measures used. Macro F1 is the headline measure for both models because it weights the 15-photograph Black Leaf Spot test class equally with the 28-photograph Phyllosticta class.

Table 3.3: Evaluation measures and what each one answers

Measure	Definition	Used for
Accuracy	share of photographs whose predicted class is correct	both models, beside the baseline
Majority-class baseline	accuracy of always predicting the largest class of the same split	interpreting accuracy under imbalance
Precision, recall, F1	per class; F1 is the harmonic mean of precision and recall	per-class behaviour
Macro and weighted F1	unweighted and support-weighted means of per-class F1	headline measure
Confusion matrix	counts of each true class against each predicted class	error analysis
Within-one-grade accuracy	share of severity grades equal to or one step from the true grade	ordered severity scale
Severe graded mild	count of the most harmful severity error	safety of severity advice
Coverage and accuracy at threshold	share answered automatically, and accuracy on those answers	choosing the confidence threshold
Refusal and false-refusal rates	share of non-orchid photographs refused; share of orchid photographs refused	input check
Model size and parameters	size of the saved model and number of weights	deployability

The improvement experiments add quadratic weighted kappa for severity [16], bootstrap confidence intervals, and the mean and spread across cross-validation folds.

3.5 Data analysis

Metrics were computed with scikit-learn from the predictions of the saved models on the held-out sets, and were written to JSON files and text reports so that every figure in Chapter 6 can be traced to a file. Every misclassified test photograph was listed with its confidence for error analysis (Appendix A), the distribution of severity grades in each split was tabulated to explain the severity results, and the distribution of feature distances was plotted for orchid and non-orchid photographs.

Small samples were treated with caution throughout. With 67 test photographs one photograph moves accuracy by 1.5 percentage points, and with 43 graded test photographs one photograph moves severity accuracy by 2.3 points. Differences of that size between models are therefore described as within noise rather than as evidence that one model is better.

3.6 Ethics privacy security and responsible AI

3.6.1 Privacy

The photographs show plants rather than people and were taken with the growers' permission. The diagnosis history is stored only on the grower's phone, and no account or personal data is required to obtain a diagnosis. A contribution records the photograph, the suggested disease and, optionally, the name of the institute or expert who confirmed it; any expansion to grower accounts would require a review under Sri Lanka's Personal Data Protection Act [22] before it was built.

3.6.2 Responsible AI

The component is designed to fail safely. An unfamiliar photograph is refused rather than diagnosed, an uncertain prediction is referred to an expert rather than reported, severity is never graded for a condition that cannot be named, and a low-confidence grade carries a note telling the grower to inspect the plant. No chemical dose is shown unless it has been verified against the product label and the registered list, because pesticide use in Sri Lanka is regulated [23] and an unsourced rate could

harm the grower, the plant or the environment. Where a dose is unverified the grower receives a referral instead.

3.6.3 Security

Uploads are checked for type, emptiness and size (12 MB) before any model is involved, and are processed in memory without temporary files. The shared backend protects the routes that control hardware with an API key; the disease routes do not change hardware, and the contribution route is being placed behind the expert review described in Section 4.4 so that no submission can influence training without an expert's approval.

3.7 Reproducibility and research artefacts

All code, saved models, metrics and reports are held in the group's public Git repository under ml-models/disease_detection, with the backend routes and mobile screens in their respective folders. The split and augmentation use a fixed seed, so rerunning them reproduces the recorded counts exactly, and the leakage report can be regenerated at any time. Training metadata records the TensorFlow version, class weights, arguments and counts for every model. The models were trained on Google Colab with TensorFlow 2.20 and confirmed to load under TensorFlow 2.21; a weights-only file is kept as a fallback in case a future version cannot read the saved model. The Colab notebook, a run guide and a component context document describe every command needed to rebuild the dataset, retrain both models and regenerate every result. Appendix F lists each artefact and the result it supports.

4 PRODUCT AND SOLUTION DESIGN

This chapter sets out the requirements of the component, its place in the complete solution, the design decisions taken and the alternatives rejected, and the detailed design of each part.

4.1 Requirements

The functional requirements (Table 4.1) follow from the objectives in Section 1.5 and from the grower's needs described in Section 1.3. The non-functional requirements (Table 4.2) are framed by the quality characteristics of ISO/IEC 25010 [24].

Table 4.1: Functional requirements

ID	Requirement	Status
FR1	Accept a leaf photograph from the phone camera or the gallery	implemented
FR2	Reject a file that is not a usable image, or larger than 12 MB, with a readable message	implemented
FR3	Refuse a photograph unlike the training photographs before any diagnosis	implemented
FR4	Classify the leaf as Black Leaf Spot, Phyllosticta Leaf Spot or healthy, with a confidence and the probability of every class	implemented
FR5	Return an unidentified condition with an expert referral when confidence is below the threshold	implemented
FR6	Grade severity only for an identified disease	implemented
FR7	Recommend treatment keyed by disease and severity, never displaying an unverified dose	implemented
FR8	Keep a history of analyses on the phone that opens without a network	implemented
FR9	Accept photographs of unknown conditions with plant part and severity, and report progress toward the training threshold	implemented
FR10	Report model status and treatment advice without loading the models	implemented
FR11	Allow an expert to approve, reject or correct each contribution, recording every decision	in progress

Table 4.2: Non-functional requirements

Characteristic	Requirement
Performance efficiency	A diagnosis returns within a few seconds once the models are loaded; models load once and are cached
Reliability	A missing model disables only the affected endpoint (HTTP 503); an unexpected label falls back to the unidentified advice instead of failing
Safety	Unverified doses are never displayed; unfamiliar or uncertain inputs are refused or referred
Usability	Each outcome is shown distinctly in plain language with a retry path on failure
Portability	The diagnosis model stays under 10 MB so that on-device deployment remains possible
Maintainability	Preprocessing is defined once, class names are saved with the model, and

Characteristic	Requirement
	treatment advice can be changed without retraining
Security	Uploads are validated before use; expert actions require authentication
Privacy	No personal data is needed for diagnosis; history stays on the phone

4.2 Component architecture within the complete solution

Figure 4.1 shows the component within the complete solution. The mobile application sends a photograph over HTTPS to the disease router in the shared FastAPI backend, which runs as a single process hosting the routers of all four components. The router validates the upload and passes it to a service module, which loads the models lazily on the first request and caches them, and which converts a missing model into a clear error on that endpoint rather than a failed server. The service calls the inference cascade, which holds every rule about when each model runs, so that the command line, the tests and the API all apply the same logic. The verdict is returned to the application and saved in the on-device history.

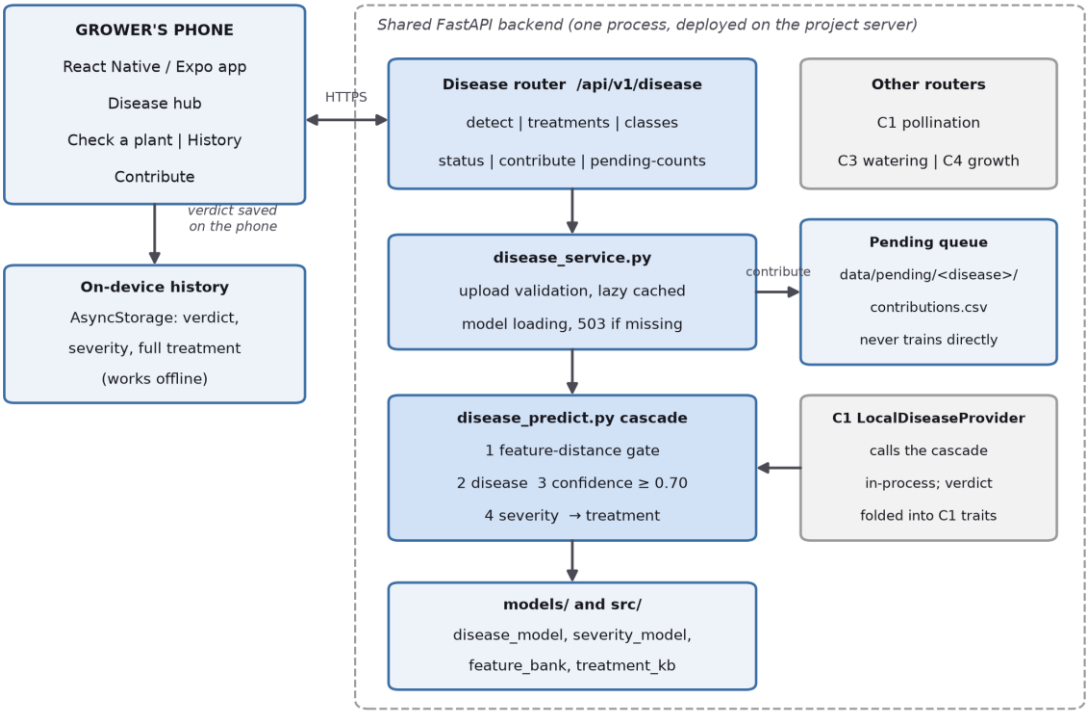


Figure 4.1: Architecture of Component 2 within the complete solution. Grey boxes belong to other components

Two other paths reach the cascade. Component 1 calls it in-process through a provider interface and folds the returned severity into its own trait vector; an unidentified or refused photograph is reported to Component 1 with zero confidence, which it must never treat as a healthy plant. Contributions of unknown conditions are written to a pending queue on the server, which never changes the deployed model directly.

4.3 Design alternatives and decisions

Table 4.3 records the main design decisions, the alternatives considered and the evidence or reasoning that decided each.

Table 4.3: Design decisions, alternatives and the reason for each choice

Decision	Alternatives	Chosen	Reason
Backbone	EfficientNetB0; network from scratch	MobileNetV2	EfficientNetB0 was better by one validation image at 1.8 times the size; the scratch network reached 0.602 macro F1
Unknown diseases	a trained 'other' class	confidence threshold plus referral	no representative 'other' data exists; an undertrained class would steal predictions
Unrelated images	softmax confidence; binary orchid classifier	feature-distance check	softmax gave 99.9% to a meal; distance needs no negative training data
Severity	two-output network; segmentation of lesion area	separate classifier	simpler to debug; no pixel-level masks were available
Augmentation	random online transforms with crop and zoom	54 fixed transforms, no crop	a crop changes the affected-area proportion that defines the grade
Background	segment the leaf before classifying	whole photograph	growers' photographs include backgrounds; segmentation would add a second model
Treatment	learned recommendation	rule-based knowledge base	advice must be sourced, inspectable and editable without retraining
History	server database with accounts	on-device storage	works offline and stores no personal data
New diseases	automatic retraining at a count	expert-verified, manually started retraining	an unverified label from one user must never reach training

4.4 Detailed design

4.4.1 Diagnosis cascade

Figure 4.2 shows the cascade that every diagnosis passes through. Each stage can end the request. Stage 1 computes the feature distance and refuses the photograph when it exceeds 0.55, returning no class and no confidence, because showing either would repeat the error the check exists to prevent. Stage 2 runs the disease classifier. Stage 3 compares the highest probability with the 0.70 threshold and returns an unidentified condition with a referral when it falls below. A confident healthy verdict ends with preventive advice and no grade. Only a confident disease verdict reaches Stage 4, where the severity grader runs and the disease and grade select the treatment. If the grade's own confidence is below 0.55, the response carries a note advising the grower that the grade may be one step out.

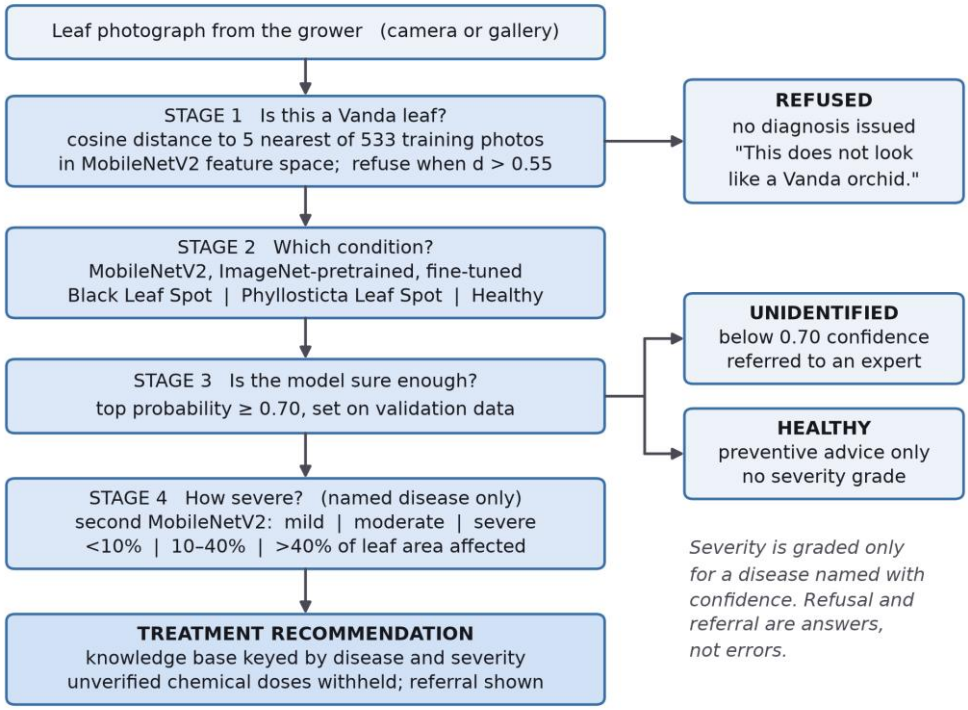


Figure 4.2: Four-stage diagnosis path. Severity is graded only for a disease identified with confidence, and refusal or referral is returned as an answer rather than an error

4.4.2 Preprocessing parity

Inference applies exactly the preparation used to build the training data: the orientation tag is applied to the pixels, the image is reduced to a longer side of 1,024 pixels with the same resampling filter, and it is then resized to 224 by 224. Before

this was enforced, photographs taken straight from a phone reached only 70.1% accuracy on the test set against 80.6% for the processed versions of the same photographs, a loss of 10.4 percentage points caused entirely by the mismatch.

4.4.3 Treatment knowledge base

Treatment advice is held in a JSON knowledge base keyed by disease and severity. Each disease records its pathogen, pathogen type and favourable conditions, and each grade records a summary, immediate actions, cultural controls, chemical options, monitoring advice and whether the case should be escalated to an expert. Each chemical option names its active ingredient and FRAC group, so that growers can rotate between groups to slow resistance, and carries a dose, a verification flag and a source reference. The advisor returns a dose only when its flag is set; otherwise it substitutes a referral message, so an unverified rate cannot reach a grower even if the application forgets to check. Separate entries cover healthy plants, unidentified conditions and refused images, and an out-of-scope section keeps the fungus-oomycete distinction for Phytophthora. Safety notes, such as reading the label and avoiding spraying in midday sun above 30 °C, are always shown. Appendix C lists every entry.

4.4.4 Application programming interface

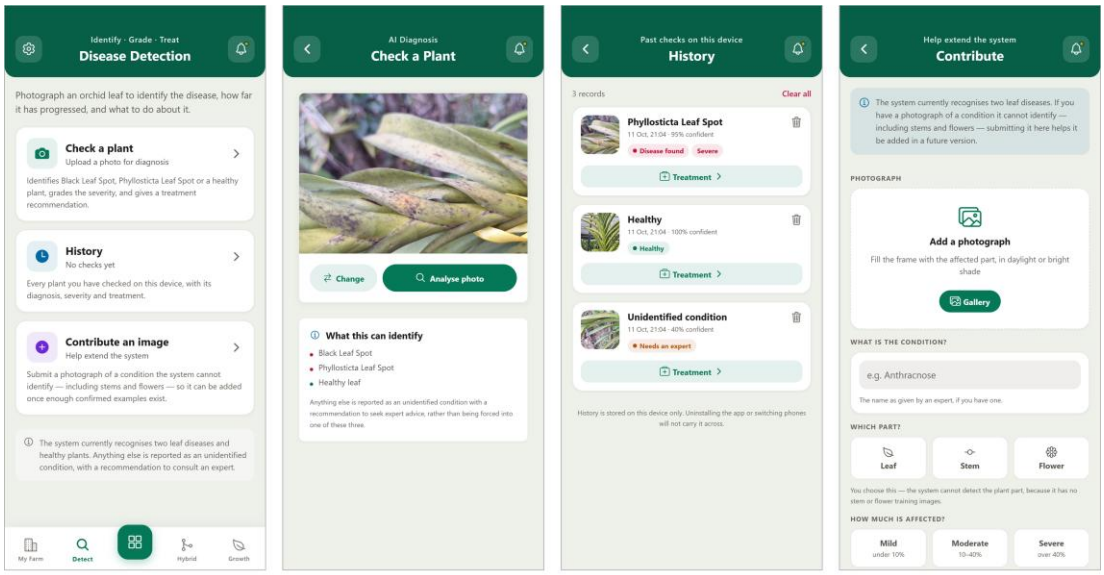
Table 4.4 lists the endpoints. Every route is thin: validation and business rules live in the service and the cascade. Appendix E gives the full request and response contracts.

Table 4.4: Endpoints of the disease router (prefix /api/v1/disease)

Method and path	Purpose	Error behaviour
POST /detect	full cascade on an uploaded photograph; optional threshold	400 invalid file; 503 model missing
GET /treatments/{disease}	advice for a disease and optional severity, without a model	an unknown name returns the unidentified advice
GET /classes	everything the system can name, with grades	503 if the knowledge base is unavailable
GET /status	which model files are present, without loading TensorFlow	reports degraded status
POST /contribute	store a photograph of an unknown condition with its labels	422 incomplete or inconsistent form
GET /pending-counts	progress of each contributed disease toward the threshold	503 on storage error

4.4.5 User interface design

The disease tab opens a hub with three tasks: check a plant, history and contribute. The detection screen accepts a photograph from the camera or the gallery and shows one of five clearly distinguished outcomes: healthy, disease found with its grade and treatment, unidentified with a referral, refused as not an orchid, or a request failure with a retry button. The history screen lists previous analyses with their verdict and grade and opens the full stored treatment without a network. The contribute screen collects a photograph, a disease name, the plant part, a severity and notes, and shows progress toward the training threshold (Figure 4.3).



(a) Hub (b) Photograph ready (c) History (d) Contribute

Figure 4.3: Disease screens of the mobile application: (a) hub, (b) a photograph ready for analysis, (c) history and (d) contribution. Captured from the application's web build at phone size, which hides the camera button; original in colour

4.4.6 Contribution and expert verification

Photographs of conditions the classifier does not recognise are stored as originals in a pending queue and are deliberately not augmented on arrival, because augmenting before a split would place variants of one leaf on both sides of it. A class may be trained only when it has at least 30 verified originals: thirty photographs augmented 54 times still carry only thirty photographs of information, and a class learned from fewer would draw a boundary from almost no evidence and steal predictions from the classes that already work. Following the panel's concern that one user could fill the counter and trigger training with unverified labels, verification is moved from the

contributor to an expert (Figure 4.4). Automatic checks reject unusable photographs and copies; an orchid researcher with an expert login approves, rejects with a reason, or corrects each photograph; only approved photographs from at least three different growers count toward the threshold, with near-copies counted once; and retraining is started manually with the same pipeline and published only if the existing classes do not get worse.

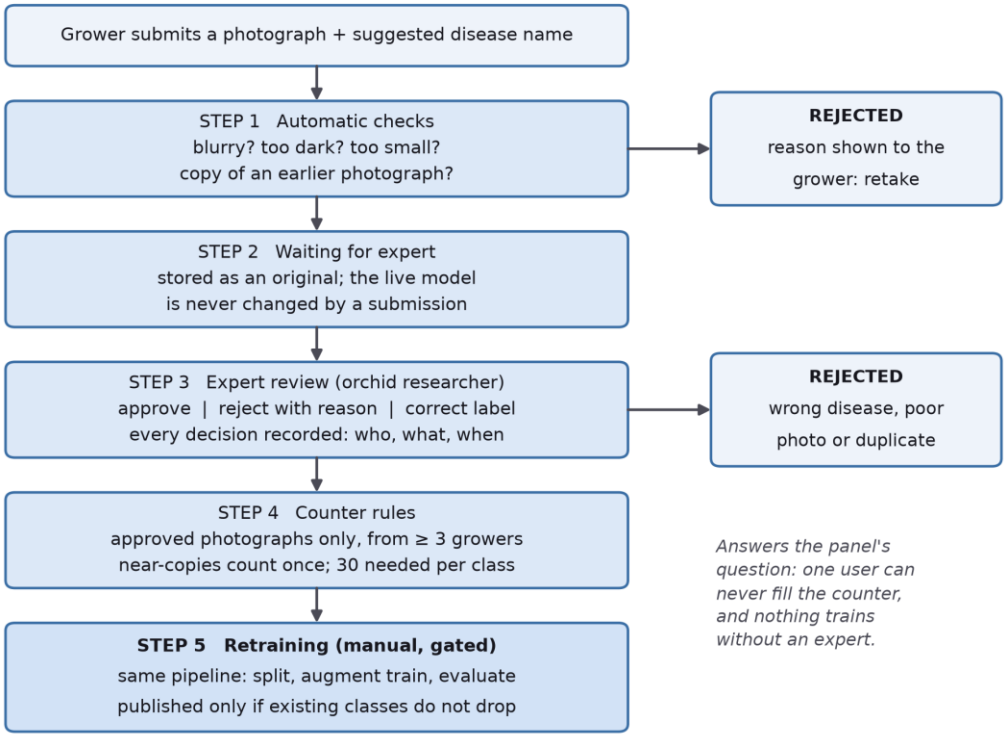


Figure 4.4: Contribution and expert verification workflow. No submission can change the deployed model, and only expert-approved photographs from several growers count toward retraining

4.5 Standards and professional practice

The non-functional requirements are organised by the ISO/IEC 25010 quality characteristics [24]. Development followed the team's Git workflow: work was done on a component branch and merged into the main branch through reviewed pull requests, and the component's documentation records every command, result and design decision. Professional responsibility was treated as a design requirement in line with the ACM Code of Ethics [25]: results are reported with their baselines and limitations, smoke-test numbers were deleted rather than left where they could be

mistaken for results, and the treatment advice withholds what cannot be sourced. References follow the IEEE style and the thesis follows the institutional formatting standards.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter describes how the design was implemented, how the component was joined to the other three, how it was verified, and how it is deployed and operated.

5.1 Development environment and technology stack

Table 5.1 lists the technologies used. The models were trained on a cloud GPU because training on a laptop CPU would have taken several hours per run, while inference, verification and the service run on a CPU.

Table 5.1: Technology stack of Component 2

Layer	Technology	Use
Model training	TensorFlow 2.20 and Keras on Google Colab (NVIDIA T4)	training both models
Local inference	Python 3.11 virtual environment, TensorFlow 2.21	evaluation, verification, cascade
Image handling	Pillow, NumPy	orientation, resizing, feature distances
Evaluation	scikit-learn, Matplotlib	metrics, confusion matrices, figures
Backend	FastAPI and Uvicorn, shared with the other components	disease router and service
Mobile application	React Native with Expo SDK 54, expo-image-picker, AsyncStorage	camera, gallery, history
Collaboration	Git and GitHub with reviewed pull requests	version control and integration

5.2 Component implementation

5.2.1 Data tools

Six command-line tools implement the preparation procedure: an orientation and integrity survey; a renaming and augmentation tool with its two modes; a generator for the severity label table that carries existing labels over and backs up the previous file whenever it is rerun, so that hours of labelling can never be lost; a labelling aid; a tool that copies severity labels into the augmentation manifests without regenerating any image; and the three-way leakage check, which exits with an error on failure so that it can gate a retrain.

5.2.2 Training and evaluation

Separate scripts train and evaluate each model and run the architecture comparison. Each training script supports a smoke-test mode that runs one epoch on a few batches to prove the code works before an hour of GPU time is spent; its numbers are never reported. Class names are written to a file the moment the datasets are built, because the index of each output is the alphabetical position of its folder and adding a class later would silently remap every prediction. A defect was found and fixed in the training script: saving the model at the end of training overwrote the best checkpoint with the in-memory weights, which after an unproductive fine-tuning stage are worse; the script now reloads the checkpoint, so the saved model, the weights fallback and the reported metrics all describe the same model.

5.2.3 Inference

The cascade module loads both models, the class names, the reference bank and the knowledge base once and caches them, because loading a model takes seconds that no web request can afford. It accepts a file path or raw bytes, so the backend passes an upload through without writing a temporary file. The module was deliberately named so that it cannot collide with the identically named prediction modules of other components (Section 5.3).

5.2.4 Backend and application

The service module validates uploads, maps missing models to HTTP 503 and invalid images to HTTP 400, and serialises contributions with a lock so that two simultaneous submissions cannot interleave rows or claim the same sequence number. Contributed photographs are stored at the same 1,024-pixel resolution as the training data, which reduces a typical 2 MB phone photograph to about 180 KB. In the application, the detection screen implements the five outcomes, the history service stores each verdict and its full treatment with AsyncStorage so that advice opens offline, and the contribute screen implements the form and progress counter.

5.2.5 Status of the expert verification

At the time of this draft the contribution route records whether a photograph was confirmed and by whom, as declared by the contributor, and stores that declaration with a source marked as self-attested so that expert-reviewed entries can later

outrank it without migrating data. The expert login, review screen, decision record and multi-grower counting rule of Figure 4.4 are being implemented [update this paragraph when the expert review is complete].

5.3 Integration with the other components

The disease router is registered in the shared backend under /api/v1/disease beside the routers of the other components, and the disease screens are registered in the shared navigation of the application, with the disease tab opening the hub. The most significant integration is with Component 1. Its pollination assessment needs to know whether a plant is diseased, and rather than train a second disease model on its own smaller dataset it calls this component's cascade in-process through a provider interface. In automatic mode the provider uses this component's model whenever it can be loaded and falls back to a colour heuristic otherwise; an unidentified or refused photograph is reported with zero confidence, which is distinct from a report of no disease.

Integration exposed a defect that no component test would have found. Four components shipped a module with the same name, and the backend places several of their source folders on the import path at once, so the disease service imported whichever module came first. It received the pollination component's module, which required an image library absent from that environment, and every diagnosis request failed with HTTP 503. Renaming the disease module removed the ambiguity. Component 3 does not yet consume the disease verdict, for example to withhold fertiliser from a severely diseased section; that link is future work.

5.4 Verification and software testing

Verification combined scripted checks of the data and models, a held-out evaluation, an end-to-end run of the deployed cascade, and tests of every endpoint against a running service (Table 5.2). The component does not yet have an automated unit-test suite; the checks below are scripted and repeatable, and adding them to an automated suite is listed as future work.

Table 5.2: Verification activities and their outcomes

Activity	Date	Outcome

Activity	Date	Outcome
Orientation and integrity survey of 667 raw photographs	29 Aug 2026	330 stored sideways and corrected; 0 corrupt; 0 duplicates
Three-way leakage check of the split	29 Aug and 10 Oct 2026	PASS: 0 overlapping sources, 0 leaked files, 0 clashes
Training smoke test, one epoch on a few batches	29 Aug 2026	code ran end to end; outputs deleted as not reportable
Knowledge base check against the model's class list	10 Oct 2026	every predictable class has advice; 12 doses flagged unverified, as expected
Held-out evaluation of both models	29 Aug 2026	results in Section 6.1
End-to-end cascade on 67 test and 17 non-orchid photographs	10 Oct 2026	results in Section 6.2
Endpoint tests against a running service	10 Oct 2026	14 of 14 behaved as designed (Table 6.7)

Table 5.3 lists the defects that verification and evaluation found and how each was fixed. Each fix is recorded in the code beside the line it changed.

Table 5.3: Defects found during development and their fixes

Defect	How it was found	Fix
Phone photographs ignored their orientation tag at inference, costing 10.4 points of accuracy	comparing processed and raw versions of the test photographs	inference applies the same orientation and resizing as training
Unrelated photographs diagnosed with high confidence (a meal as healthy at 99.9%)	testing with non-orchid photographs	feature-distance input check
Disease requests failed with HTTP 503 after integration	running the shared backend	unique module name for the cascade
Best checkpoint overwritten when training finished	comparing saved and logged metrics	reload the checkpoint before saving
Rotated training images had black corners	inspecting augmented files	crop the largest clean rectangle
One photograph sorted first and shifted every Black Leaf Spot number by one	tracing processed files to raw files	renamed so raw and processed numbers match

5.5 Deployment and operation

The component is deployed as part of the shared backend on the project's virtual private server, reached through a domain name, and the application reads the server address from one configuration module. The saved models are held in the repository and loaded on the first request, and the status endpoint reports which files are present without loading TensorFlow, so a partly provisioned server is visible rather than silent. Contributed photographs are stored on the server's disk and are deliberately

excluded from version control; they must therefore be backed up separately [state the backup arrangement].

Retraining is a manual, documented procedure rather than an automatic one. When a contributed class reaches its threshold of approved photographs, the approved originals are exported as a new class folder and passed through the existing pipeline: split, augmentation of the training portion only, leakage check, training and evaluation. A comparison of the new and current evaluation results decides whether the new model is published; previous models are kept so that a release can be rolled back. [Confirm that the latest version of the component has been merged and deployed before submission.]

6 RESULTS AND VALIDATION

This chapter reports the measured results. Every figure comes from the saved evaluation files or from runs dated in Table 5.2; none was typed in by hand.

6.1 Research results

6.1.1 Disease classifier

On the 67 held-out test photographs the classifier reaches 0.806 accuracy (54 of 67), 0.778 macro F1 and 0.803 weighted F1, against a majority-class baseline of 0.418, the share of Phyllosticta photographs in the test set. On the validation split it reaches 0.791 accuracy and 0.769 macro F1, so the test result is not an outlier. Table 6.1 gives the per-class results and Figure 6.1 the confusion matrix.

Table 6.1: Per-class results of the disease classifier on the held-out test set

Class	Precision	Recall	F1	Support
Black Leaf Spot	0.600	0.600	0.600	15
Healthy	0.889	1.000	0.941	24
Phyllosticta Leaf Spot	0.840	0.750	0.792	28
Macro F1			0.778	67
Weighted F1			0.803	67
Accuracy (baseline 0.418)			0.806	67

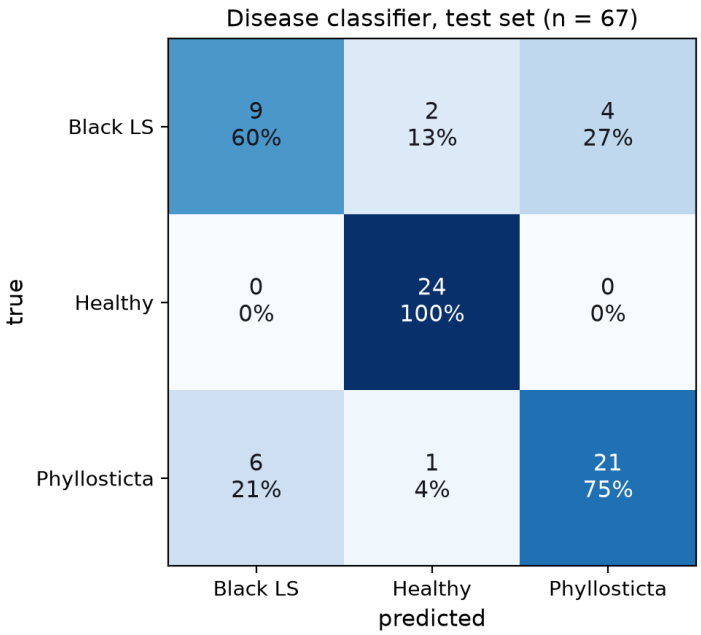


Figure 6.1: Confusion matrix of the disease classifier on the held-out test set, with counts and row percentages

Four observations follow. All 24 healthy test leaves were recognised as healthy, and no healthy leaf was called diseased, so the model never sends a grower to spray a healthy plant. Ten of the thirteen errors are confusions between the two leaf-spot diseases, which look alike. Black Leaf Spot is the weakest class (F1 0.60) and also the smallest, with 122 training originals against 219 for Phyllosticta. Three of the 43 diseased test leaves were called healthy, which is the costly direction of error, and one of them received 0.995 confidence, an error that no confidence threshold can catch.

6.1.2 Training behaviour

Figure 6.2 shows accuracy and loss over the 21 training epochs. Validation accuracy plateaued at about 0.82 with the network frozen. Fine-tuning raised training accuracy to 0.98 and validation accuracy to 0.88 by epoch 21, but validation loss rose at the same time, a sign of growing over-confidence. Because checkpoints were selected by validation loss, the saved model is the one from the first fine-tuning epoch, whose validation accuracy is 0.791; this was confirmed by re-measuring the saved model's validation loss and by comparing its weights with the original ImageNet weights, which differ only in the top inverted-residual blocks. Selecting checkpoints by validation macro F1 is one of the improvement experiments.

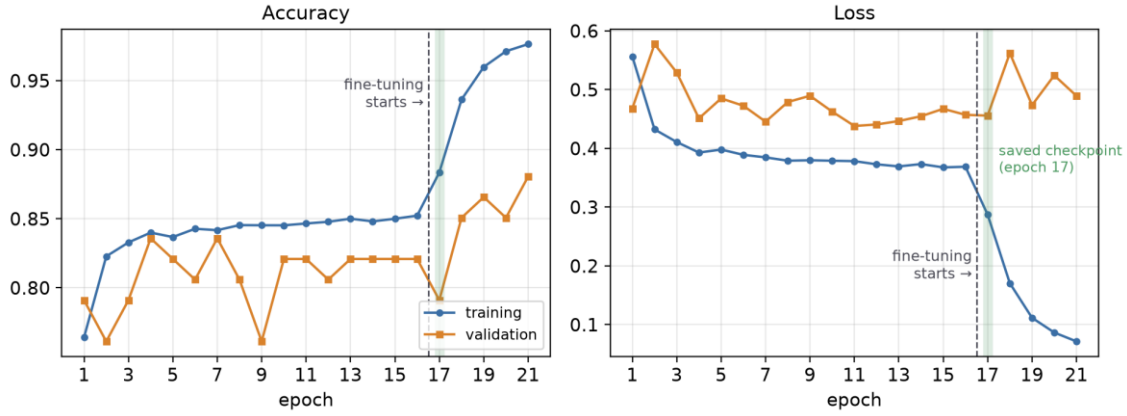


Figure 6.2: Training and validation accuracy and loss of the disease classifier. The dashed line marks the start of fine-tuning and the shaded band the saved checkpoint

6.1.3 Backbone comparison

Under identical conditions on the validation split, EfficientNetB0 reached 0.832 macro F1 and MobileNetV2 0.822, while the network trained from scratch reached

only 0.602 (Table 6.2 and Figure 6.3). The scratch network's shortfall shows that 533 original photographs cannot teach useful convolutional filters from nothing, so transfer learning was necessary. EfficientNetB0's advantage amounts to a single validation photograph in accuracy, within the noise of a 67-photograph set, at 1.8 times the model size, so MobileNetV2 was selected. The comparison also recorded latency, but it was measured on a cloud GPU, where MobileNetV2's depthwise convolutions are less efficient than on the phone processors they were designed for; the decision therefore rests on model size, and latency on the target device was not measured.

Table 6.2: Backbone comparison on the validation split (five epochs, frozen pretrained layers)

Model	Macro F1	Accuracy	Parameters	Size (MB)
EfficientNetB0	0.832	0.851	4,053,414	17.1
MobileNetV2 (selected)	0.822	0.836	2,261,827	9.7
CNN trained from scratch	0.602	0.746	94,531	1.2

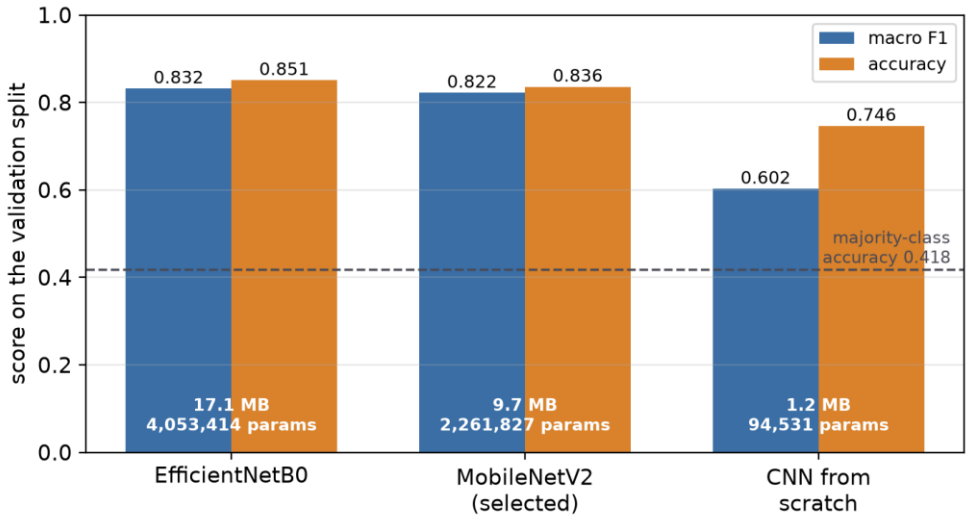


Figure 6.3: Backbone comparison on the validation split, with model size and parameter count

6.1.4 Confidence threshold

Raising the threshold refers more photographs to an expert and makes the remaining answers more accurate (Figure 6.4 and Table 6.3). At 0.70 the system answers 83.6% of validation photographs automatically at 91.1% accuracy, against 79.1% when every photograph is answered, and it behaves consistently on the test split, answering 85.1% at 86.0% accuracy. Every photograph in these sweeps belongs to a trained

class, so the sweep measures the cost of referring known diseases; the benefit of referring a disease the model has never seen could not be measured, because no photographs of such a disease were available.

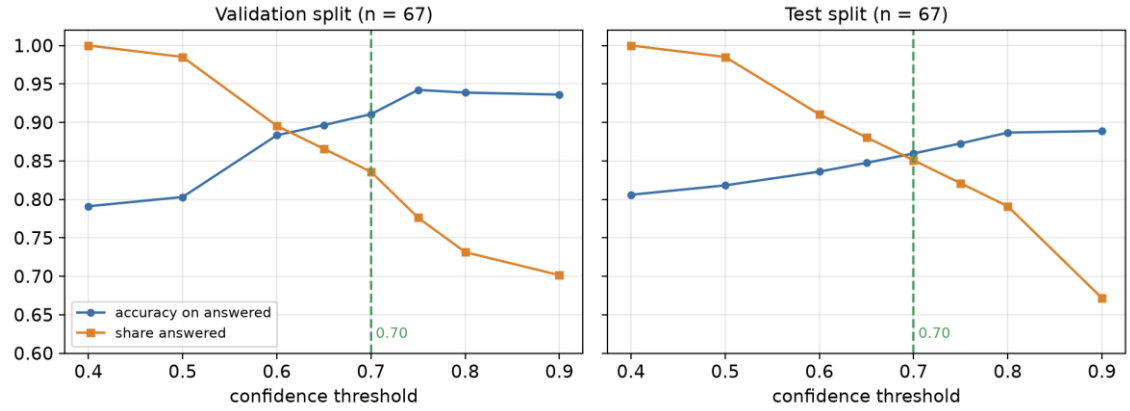


Figure 6.4: Share of photographs answered automatically and accuracy on those answers, by confidence threshold. The threshold of 0.70 was chosen on the validation split

Table 6.3: Effect of the confidence threshold on the validation and test splits

Threshold	Validation answered	Validation accuracy on answered	Test answered	Test accuracy on answered
0.50	98.5%	0.803	98.5%	0.818
0.60	89.6%	0.883	91.0%	0.836
0.65	86.6%	0.897	88.1%	0.847
0.70 (chosen)	83.6%	0.911	85.1%	0.860
0.75	77.6%	0.942	82.1%	0.873
0.80	73.1%	0.939	79.1%	0.887

6.1.5 Severity grader

On the 43 graded test photographs the severity grader reaches 0.465 exact accuracy and 0.450 macro F1, against a majority-class baseline of 0.395 and a chance level of 0.333; 90.7% of its grades are within one step of the true grade (Table 6.4 and Figure 6.5). Of 23 errors, 19 are one grade out and 4 are two grades out. The most harmful error, a severe leaf graded mild, which would lead a grower to under-treat, occurred in 2 of 14 severe cases. Moderate is the weakest grade, because as the middle of an ordered scale it can be confused with neighbours on both sides: of 12 moderate leaves, 6 were graded mild and 3 severe.

Table 6.4: Per-grade results of the severity grader on the held-out test set

Grade	Precision	Recall	F1	Support

Grade	Precision	Recall	F1	Support
mild	0.556	0.588	0.571	17
moderate	0.231	0.250	0.240	12
severe	0.583	0.500	0.538	14
Macro F1			0.450	43
Exact accuracy (baseline 0.395)			0.465	43
Within-one-grade accuracy			0.907	43
Severe graded mild			2	14

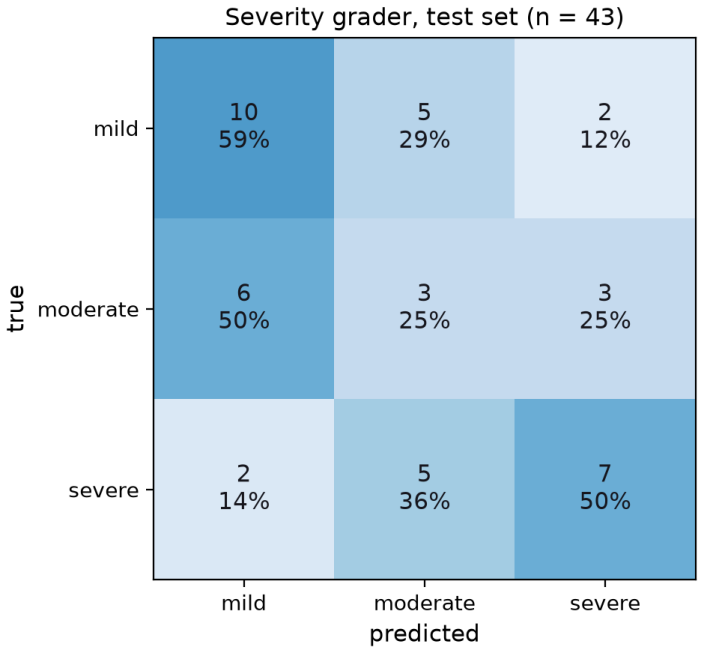


Figure 6.5: Confusion matrix of the severity grader on the held-out test set

The severity result has an identified cause. The photographs were split by disease before severity labels existed, so the grade mix of each split was left to chance, and it came out skewed: severe leaves make up 45% of the training split but 21% of the validation split (Table 6.5). The grader was also trained with its pretrained layers frozen. A split stratified by both disease and severity, and fine-tuning, are both part of the improvement experiments.

Table 6.5: Distribution of severity grades in each split (diseased originals)

Split	Mild	Moderate	Severe	Total
Train	74 (22%)	112 (33%)	155 (45%)	341
Validation	18 (42%)	16 (37%)	9 (21%)	43
Test	17 (40%)	12 (28%)	14 (33%)	43

6.1.6 Input check

Figure 6.6 plots the feature distance of every held-out test photograph and every non-orchid photograph. At the threshold of 0.55 no test photograph is refused (the largest distance is 0.453) and 15 of the 17 non-orchid photographs are refused. The two that pass are a photograph of grass (0.469) and of a tree (0.515), both green plants, which is the hardest boundary. The eleven web photographs of genuine diseased orchids, taken by other people on other equipment, lie between 0.313 and 0.547, so their range overlaps the non-orchid range and no threshold separates the two groups perfectly. A tighter threshold of 0.465 would have refused all 17 non-orchid photographs, but it also refused 2 of 145 genuine orchid photographs and sat only 0.004 below the closest non-orchid image, a margin too narrow to rely on. Refusing a genuinely diseased plant is the more harmful error for a grower, so 0.55 was chosen, accepting all 145 orchid photographs. By contrast, maximum softmax probability could not reject these images at all: the meal, grass and tree photographs described in Section 1.3 all received confidences above 99%. Appendix D lists the distance and the decision for every non-orchid photograph.

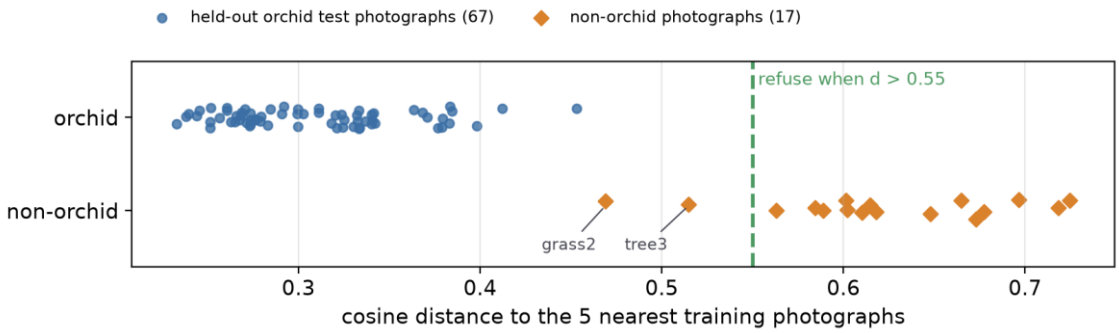


Figure 6.6: Feature distance of held-out orchid test photographs and non-orchid photographs. Photographs to the right of the dashed line are refused

6.1.7 Improvement experiments

[Insert Table 6.x with one row per experiment of Section 3.3: the change made, validation accuracy and macro F1, and whether the change was kept. Report the final models' test results and the cross-validation mean and spread here, and state any change that did not help.]

6.2 Product and technical results

6.2.1 End-to-end behaviour

On 10 October 2026 the deployed cascade, the same function the backend calls, was run over all 67 test photographs and all 17 non-orchid photographs (Table 6.6). It reproduced the evaluated accuracy of 0.806 exactly, which confirms that the deployed preprocessing matches the evaluated one, and it enforced every safety rule of the design.

Table 6.6: End-to-end run of the deployed cascade, 10 October 2026

Measure	Result
Held-out test photographs processed	67
Refused by the input check	0 (largest distance 0.453)
Accuracy of the top prediction, all photographs	0.806
Answered at the 0.70 threshold	58 of 67 (86.6%)
Accuracy on answered photographs	0.845
Returned as unidentified and referred	9
Outcomes	healthy 27, Phyllosticta 20, Black Leaf Spot 11, unidentified 9
Severity graded	31 photographs; 0 of them healthy
Chemical dose displayed in any response	no
Non-orchid photographs refused	15 of 17
Non-orchid photographs accepted	grass2 (0.469), tree3 (0.515)
Time for all 84 photographs, laptop CPU	22.7 s including model loading

At the 0.70 threshold the cascade answered 58 test photographs, one more than the offline sweep's 57, because the deployed path decodes images with a different library and a single photograph lies within rounding distance of the threshold. The 84 photographs were processed in 22.7 seconds on a laptop CPU, including the one-off loading of both models.

6.2.2 Endpoint behaviour

Fourteen requests covering every endpoint and every error path were sent to a running instance of the service, with contributions redirected to a temporary folder (Table 6.7). Every request returned the designed status and content: valid photographs were diagnosed, a non-orchid photograph was refused, invalid files returned HTTP 400, incomplete contributions returned HTTP 422, and an unknown disease name fell back to the unidentified advice.

Table 6.7: Endpoint tests against a running service, 10 October 2026

Test	Request	Expected	Observed

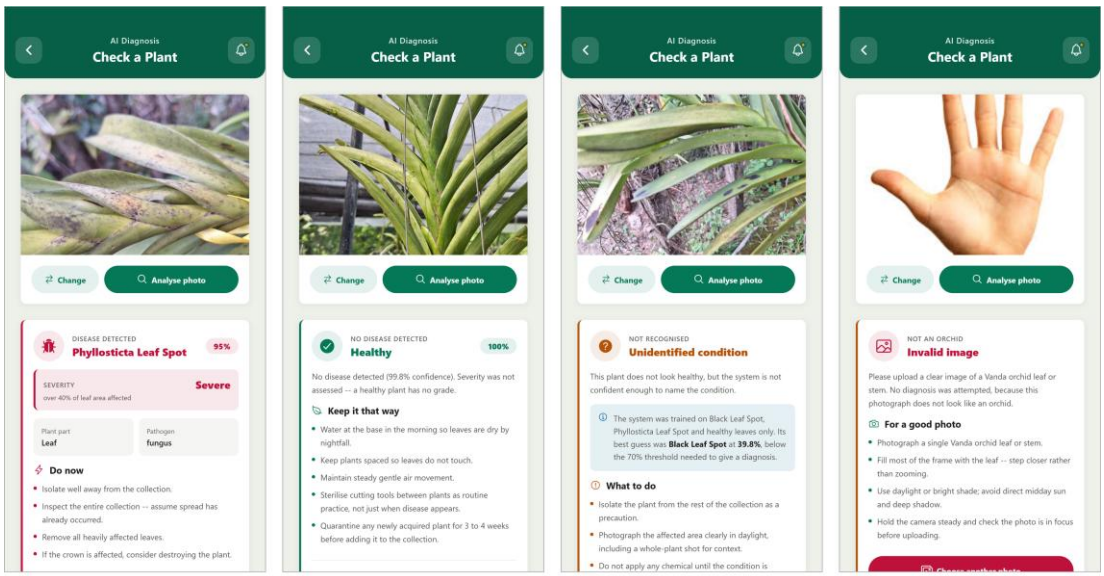
Test	Request	Expected	Observed
T1	GET /status	200, all model files present	200, all present
T2	GET /classes	200, trained classes and grades	200
T3	GET /treatments/black_leaf_spot?severity=moderate	200, advice for the grade	200
T4	GET /treatments/not_a_disease	200, unidentified advice	200, unidentified advice
T5	POST /detect, healthy test photograph	200, healthy	200, healthy at 0.998
T6	POST /detect, Phyllosticta test photograph	200, a diagnosis	200, Black Leaf Spot at 0.967 (a recorded test error)
T7	POST /detect, photograph of a dog	200, refused	200, refused as not an orchid
T8	POST /detect, text file	400	400, unsupported file type
T9	POST /detect, empty file	400	400, file is empty
T10	POST /detect, corrupt file labelled JPEG	400	400, could not read the image
T11	POST /contribute, confirmed but no verifier named	422	422, verifier required
T12	POST /contribute, invalid plant part	422	422, plant part must be leaf, stem or flower
T13	POST /contribute, complete and unconfirmed	200, saved, counts 0 of 30	200, saved, 0 of 30
T14	GET /pending-counts	200, progress per disease	200, 1 submission, 0 verified

6.2.3 Outcomes in the application

A further run of eight Phyllosticta test photographs through the service illustrates the outcomes a grower sees: five were diagnosed correctly with a grade and four immediate actions, two were the recorded test errors, and one, with a highest probability of 0.398, was returned as unidentified with no grade. No response in any run displayed a chemical dose.

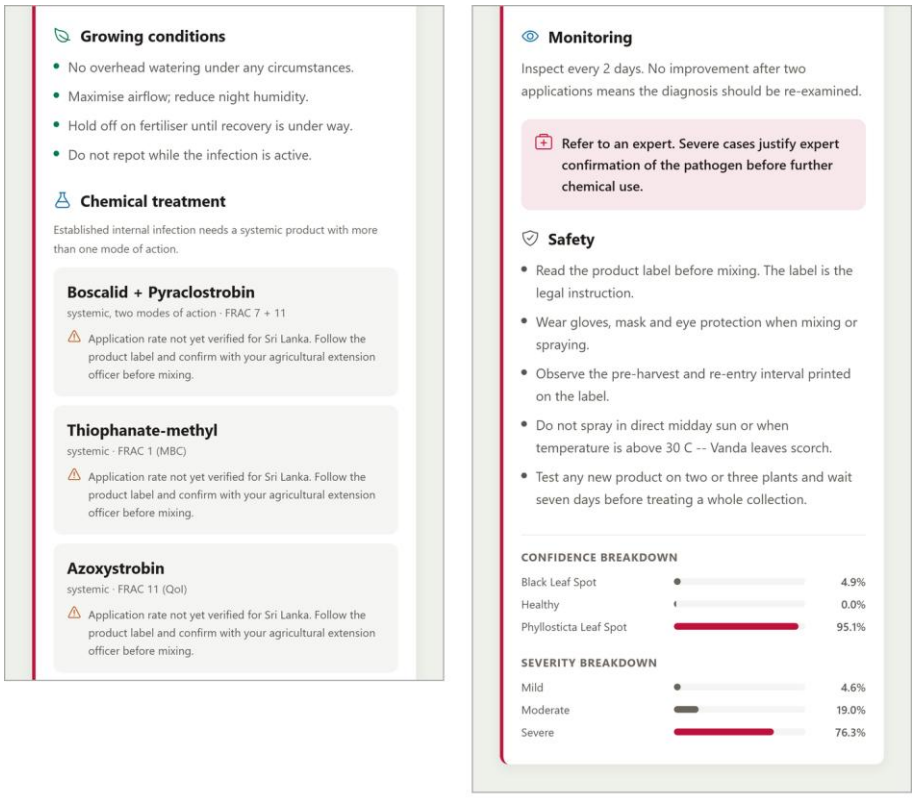
Figure 6.7 shows the four outcomes as a grower sees them, captured from the running system with photographs used in the tests above: a diseased leaf identified as Phyllosticta Leaf Spot at 95% and graded severe; a healthy leaf with preventive advice and no grade; a diseased leaf whose best guess, Black Leaf Spot at 39.8%, fell below the threshold and was referred to an expert without a grade; and a photograph of a hand, refused before any diagnosis. Figure 6.8 shows the rest of the severe case: each chemical option is named with its FRAC group, every unverified rate is

replaced by a referral to the product label and an extension officer, the case is escalated to an expert, and the safety notes and probability breakdowns are shown.



(a) Disease, graded (b) Healthy (c) Unidentified (d) Refused

Figure 6.7: The four outcomes shown to a grower: (a) disease identified and graded, (b) healthy, (c) unidentified and referred, (d) refused as not an orchid. Captured from the application's web build at phone size; original in colour



(a) Conditions and chemicals (b) Referral, safety, probabilities

Figure 6.8: Treatment advice for the severe Phyllosticta case of Figure 6.7(a): (a) growing conditions and chemical options with unverified rates withheld, (b) expert referral, safety notes and probability breakdowns. Original in colour

6.3 User or stakeholder validation

[Report the expert session here: date, the researcher's name, role and institute, and what was reviewed (the disease classes, a sample of severity grades, every treatment entry and the chemical doses, and contributed photographs through the review screen). State the findings, the changes made in response, any dose now verified with its source, and any request that was not acted on with the reason.]

[Report grower feedback here: number of growers, how they used the application, and their principal comments.]

At the time of this draft, formal stakeholder validation of Component 2 had not yet been completed, and no result in this thesis depends on it.

6.4 Integration results

The component runs inside the shared backend alongside the other three routers, and its screens are reached from the shared application. Component 1 consumes its classifier at run time through the provider interface, as documented in the common integrated solution report, and the defect that made every diagnosis fail after

integration was found and fixed (Section 5.3). Component 3 does not yet consume the disease verdict.

6.5 Summary of findings

- RQ1: the classifier reaches 0.806 accuracy and 0.778 macro F1 on held-out field photographs against a 0.418 baseline; healthy leaves are recognised without error and the dominant error is between the two leaf spots.
- RQ2: transfer learning is necessary (0.602 from scratch against 0.822); MobileNetV2 matches EfficientNetB0 within one validation photograph at 57% of its size.
- RQ3: severity reaches 0.465 exact accuracy against a 0.395 baseline but 90.7% within one grade; most errors are one grade out, and a skewed split and a frozen network are identified causes.
- RQ4: a feature distance refused 15 of 17 non-orchid photographs while accepting every genuine orchid photograph tested, where softmax confidence exceeded 99% for unrelated images.
- RQ5: a 0.70 threshold answers about 84% of photographs automatically at about 91% validation accuracy.

7 DISCUSSION

This chapter interprets the results, compares them with the literature and existing systems, states the research and product contributions, and sets out the limitations and implications.

7.1 Interpretation of findings

The disease classifier is useful in the way that matters most to a grower: it never told a grower to spray a healthy plant, and when it errs it usually confuses two leaf spots that are treated with overlapping products. Its weakest class is also its smallest, which points to more Black Leaf Spot photographs rather than a different architecture as the most direct improvement. The confident error on a diseased leaf called healthy is a reminder that a threshold cannot catch every mistake; the history and referral paths exist so that a grower who doubts a verdict can seek a second opinion.

The training curves show that fine-tuning was helping accuracy while making the network over-confident, and that selection by validation loss kept an early checkpoint. This is a measurable opportunity rather than a defect, and with only 67 validation photographs any gain must be confirmed on the test split and across folds before it is claimed.

Severity is the component's weakest result, and the reasons are understood. The grades are cut-points on a continuous quantity, so a leaf at 9% affected area and one at 11% look almost identical but carry different labels; nineteen of the twenty-three errors are of exactly that kind. The middle grade is structurally the hardest, and the skewed split means the grader trained mostly on severe leaves was validated mostly on mild ones. Within-one-grade accuracy of 90.7% means the grade is a usable indication but not a measurement, and the application says so whenever the grade is uncertain.

The input check shows that refusal can be measured rather than asserted, and that it has a cost. The chosen threshold refuses nothing genuine in the photographs tested but lets two green plants through; the alternative would have refused real orchids

photographed by other people. The trade-off was decided by the consequence of each error, but it rests on only 17 non-orchid photographs, and a larger benchmark is needed before the refusal rate can be stated with confidence.

Finally, the 10.4-point loss caused by a preprocessing mismatch is a deployment lesson in its own right: a model evaluated on carefully prepared images can perform markedly worse on the raw photographs a phone produces, and only a deliberate comparison between the two revealed it.

7.2 Comparison with literature baselines and existing solutions

Table 7.1 places this component's result beside the reviewed results. The figures are not directly comparable: they cover different crops, classes, image sources and evaluation conditions, and some report detection rather than classification. What the comparison shows is that the reviewed figures were obtained under conditions that the general plant disease literature has found to overstate field performance [4], [20], [21], whereas this component's figure is held-out, leakage-checked and stated beside its baseline. A two-class problem also has a much higher chance level than a three-class one.

Table 7.1: Reported results of the reviewed work beside this component

Work	Task	Reported result	Evaluation conditions stated
Santisarn and Titijaroonroj [1]	orchid leaf, 2 classes	98% accuracy	no baseline reported
Tsai et al. [2]	orchid viral diseases	0.918 identification accuracy	no baseline reported
OrchiZen [3]	fungal disease detection, 3 genera	detection rates 87% to 91%	no held-out split or baseline described
Mohanty et al. [4]	38 crop-disease classes	99.35% on same-source images	just above 31% on other sources
This component	Vanda leaf, 3 classes	0.806 accuracy, 0.778 macro F1	held-out originals, leakage-checked, baseline 0.418

Against existing systems, the component's distinguishing features are not accuracy in the abstract. It grades severity, which none of the reviewed orchid systems does; it ties treatment to disease and grade while withholding unverified doses, where OrchiZen lists fungicides per disease [3]; it refuses unrelated photographs; and it provides an expert-verified route for adding diseases. Its weaknesses relative to them

are equally clear: it covers fewer diseases than OrchiZen, which also treats infection physically with UV-C light, and it does not anticipate outbreaks from environmental signals as the sensor-based systems do [6], [19].

7.3 Research contribution

- A held-out, leakage-checked evaluation of leaf-disease diagnosis on field photographs of Vanda, reported beside its majority-class baseline.
- A measured demonstration that transfer learning is necessary at this data scale, through a control network trained from scratch.
- A characterisation of severity grading on an ordered scale, separating exact and within-one-grade accuracy and identifying the middle-grade and split-stratification effects.
- A measured demonstration that softmax confidence does not reject unrelated photographs, and a feature-distance check that does, with its cost stated.
- A measured estimate of the accuracy lost when inference preprocessing differs from training preprocessing.

7.4 Product contribution

The product contribution is a working diagnosis service and mobile interface: camera and gallery input, five clearly separated outcomes, treatment advice that cannot display an unverified dose, an offline history, a contribution queue, an endpoint that reports model status for operations, and an in-process interface through which another component consumes the diagnosis. Each is backed by scripts and documentation that let another developer rebuild the dataset, retrain the models and regenerate every result.

7.5 Limitations and threats to validity

7.5.1 Internal validity

The test sets are small, at 67 photographs for disease and 43 for severity, so differences of a few percentage points are within noise. Results come from a single split rather than cross-validation. Model choices were made on a 67-photograph

validation set. The architecture comparison used five epochs with frozen networks, and the checkpoint of the deployed classifier was selected by validation loss.

7.5.2 Construct validity

Severity grades were assigned by one person by estimating affected area against fixed bands, and no second rater has yet measured their agreement. The treatment advice has not yet been reviewed by an expert, and every chemical dose remains unverified.

7.5.3 External validity

The photographs come from [state the sites] and may not represent every growing region, camera or lighting condition. Only two diseases are classified, and the benefit of the confidence threshold for diseases outside them is unmeasured. The input check was calibrated on 17 non-orchid photographs and lets some green plants through. The contribution workflow's verification was self-declared until the expert review is completed.

7.6 Implications

For growers, the component is decision support rather than a replacement for expertise: it triages, grades and advises, and it says when to ask an expert. For researchers, the results support reporting every figure beside its baseline, stating how images were separated, and evaluating ordered outputs with ordinal measures. For practitioners, they show that refusal must be designed and measured, and that training and inference preprocessing must match. For agricultural advisory practice, they show that automated treatment advice is only as safe as its sources, which is why the label and the registered list remain the authority.

8 COMMERCIALISATION AND SUSTAINABILITY

This chapter considers who the component serves, how it compares with alternatives, how it could be sustained, and its contribution to the Sustainable Development Goals.

8.1 Value proposition and target users

The primary users are smallholder Vanda growers, for whom the component offers an immediate first diagnosis at the bench, a severity grade that indicates how urgently to act, and advice that is safe because it never invents a chemical rate. Secondary users are agricultural extension officers, who could use the diagnosis history to advise several growers remotely, nurseries managing larger collections, and orchid research institutes, for whom verified contributions become a growing, labelled record of disease in the field.

8.2 Market and competitive position

The alternatives are general plant identification applications, which are free but identify species rather than assess disease and answer regardless of what they are shown; research systems such as OrchiZen [3], which cover more tasks and add physical UV-C treatment but require dedicated hardware; chatbot classifiers such as the Thai system reviewed in Chapter 2 [1], which classify a single disease; and extension services, which provide expertise but not at the moment of decision. The component's position is a free, phone-only diagnosis for Vanda that grades severity, refuses what it cannot interpret and withholds unverified doses. [Add named competing products and their pricing where public.]

8.3 Business or sustainability model

Consistent with the group's model of a free application with optional paid sensor hardware for irrigation, photograph-based diagnosis would be offered free, because the growers who would benefit most are least able to pay for software whose value they have not yet seen. Sustaining it requires two recurring inputs: server capacity, which is modest because inference runs on a CPU, and expert time to review contributions and verify treatment entries. A partnership with an orchid research

institute or the Department of Agriculture, in which the partner reviews contributions in exchange for access to the verified field data, is the most plausible way to provide the second. No revenue has been realised and no pricing has been tested with growers.

8.4 Scalability intellectual property and compliance

Inference on a CPU server scales by adding capacity, and the 9.7 MB diagnosis model is small enough to be moved onto the phone in future, which would remove the network dependency. New diseases are added through the verified contribution route rather than by redesign. The photographs were collected with growers' permission and their reuse beyond the project should be agreed with them. The software dependencies are published under permissive open-source licences. Compliance rests on two points: chemical advice must follow registered product labels under Sri Lanka's pesticide regulation [23], and any move to grower accounts must comply with the Personal Data Protection Act [22].

8.5 Contribution to the Sustainable Development Goals

The most direct contribution is to Goal 12, responsible consumption and production, through target 12.4 on the sound management of chemicals. The component never recommends spraying a healthy plant, scales its advice to severity so that a mild case receives sanitation and cultural control before chemical options, and withholds unverified doses. A secondary contribution is to Goal 8 through target 8.2, productivity through technological upgrading, by making diagnostic judgement available to less experienced growers. Neither effect has yet been measured in a working shade house, and both are stated as plausible pathways rather than demonstrated outcomes.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis set out to turn a phone photograph of a Vanda leaf into a trustworthy diagnosis, severity grade and treatment recommendation from a few hundred field photographs, while refusing what the system cannot interpret. The component achieves this for the two diseases it covers. Its classifier reaches 0.806 accuracy and 0.778 macro F1 on held-out photographs against a 0.418 baseline, under an evaluation that cannot have leaked. Its cascade refuses unrelated photographs, refers uncertain ones, grades only what it can name and never shows an unverified dose. Its severity grader is weak in exact terms but usable as an indication, and the causes of its weakness are identified. The component is deployed in the shared system and is consumed by another component at run time.

9.2 Achievement of objectives

Table 9.1 summarises the evidence for each objective set in Section 1.5 and states how far it was achieved.

Table 9.1: Achievement of the objectives

Objective	Evidence	Outcome
O1 Leakage-free dataset	Figure 3.2; leakage check passed twice	Achieved
O2 Disease classifier	Tables 6.1 and 6.2; Figure 6.3	Achieved
O3 Severity grader	Table 6.4; 90.7% within one grade	Partly achieved; improvement experiments under way
O4 Input check and threshold	Figure 6.6; Table 6.3	Achieved; larger non-orchid benchmark needed
O5 Treatment knowledge base	Section 4.4; no dose displayed in any run	Achieved for safety; doses await verification
O6 Deployment, integration and verified contributions	Sections 5.3 and 5.5; Table 6.7	Deployment and integration achieved; expert review in progress

9.3 Individual research and product contributions

The individual research contribution is a held-out, leakage-checked evaluation of Vanda leaf-disease diagnosis with its baseline, a characterisation of ordinal severity grading, and measured evidence on transfer learning, input refusal and preprocessing parity. The individual product contribution is the dataset pipeline, the two models,

the inference cascade with its input check, the treatment knowledge base with its safety rule, and the backend service and mobile screens through which growers and Component 1 use them.

9.4 Future directions

- Complete the improvement experiments of Section 3.3 and report cross-validated results with confidence intervals.
- Re-split the data stratified by disease and severity, fine-tune the severity grader, and train it with a loss that respects the order of the grades.
- Measure agreement between the author's severity grades and an expert's on a sample of photographs.
- Build a larger benchmark of mistaken uploads, including blurred, dark and partial photographs and other plants, and add image-quality checks that tell the grower how to retake the photograph.
- Complete the expert review of contributions and verify every chemical dose against the label and the registered list.
- Show the grower which part of the leaf drove the diagnosis with gradient-based heat maps [26].
- Move the diagnosis model onto the phone so that it works without a network.
- Let Component 3 act on the disease verdict, for example by withholding fertiliser from a severely diseased section.
- Collect more Black Leaf Spot photographs and add the deferred diseases as verified contributions accumulate.

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GLOSSARY

Augmentation - creating modified copies of training photographs (rotated or recoloured) so that a model learns that a lesion is the same lesion under different conditions.

Baseline (majority-class) - the accuracy obtained by always predicting the most common class of a split; a model must exceed it to show that it has learned anything.

Checkpoint - a saved copy of a model's weights at a particular point in training.

Confusion matrix - a table counting, for each true class, how often each class was predicted.

Cosine distance - one minus the cosine of the angle between two feature vectors; small values mean the two images look alike to the network.

Data leakage - information about the test set reaching training, which makes reported results look better than real performance.

Epoch - one complete pass of the training data through the model.

Fine-tuning - continuing to train some layers of a pretrained network on a new task at a small learning rate.

Held-out test set - photographs kept aside and used only once, to measure the final model.

Macro F1 - the unweighted mean of the F1 scores of all classes, so that small classes count as much as large ones.

Oomycete - a fungus-like organism, such as Phytophthora, that responds to different chemistry from true fungi.

Out-of-distribution input - an input unlike anything the model was trained on.

Softmax - the final layer of a classifier, which turns scores into probabilities that sum to one across the trained classes.

Transfer learning - reusing a network trained on a large dataset as the starting point for a new task with less data.

Within-one-grade accuracy - the share of severity predictions that are correct or one grade away from correct.

APPENDICES

Appendix A Extended results

Table A.1 lists every misclassified photograph in the disease classifier's test set with the confidence of the wrong prediction. Three of the thirteen are diseased leaves predicted healthy, the costly direction of error.

Table A.1: Misclassified test photographs of the disease classifier

Photograph	True class	Predicted class	Confidence
Black_LS_0007.jpg	Black Leaf Spot	Healthy	0.856
Black_LS_0008.jpg	Black Leaf Spot	Healthy	0.995
Black_LS_0009.jpg	Black Leaf Spot	Phyllosticta Leaf Spot	0.586
Black_LS_0023.jpg	Black Leaf Spot	Phyllosticta Leaf Spot	0.692
Black_LS_0027.jpg	Black Leaf Spot	Phyllosticta Leaf Spot	0.781
Black_LS_0029.jpg	Black Leaf Spot	Phyllosticta Leaf Spot	0.642
Phyllosticta_LS_0002.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.732
Phyllosticta_LS_0014.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.968
Phyllosticta_LS_0041.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.428
Phyllosticta_LS_0072.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.520
Phyllosticta_LS_0102.jpg	Phyllosticta Leaf Spot	Healthy	0.972
Phyllosticta_LS_0112.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.914
Phyllosticta_LS_0141.jpg	Phyllosticta Leaf Spot	Black Leaf Spot	0.944

Every result in Chapter 6 can be regenerated from the repository with the commands documented in the component's run guide: the preparation tools in tools/, the training and evaluation scripts in src/, and the Colab notebook for GPU training.

Appendix B Severity grading protocol

Every one of the 427 diseased original photographs was graded before any severity model was trained; the 240 healthy photographs carry no grade. Grades were entered in a label table with one row per original photograph and were inherited by its augmented copies through the augmentation manifests, which is valid only because no augmentation changes the proportion of diseased tissue. Table B.1 gives the bands and the number of photographs in each.

Table B.1: Severity grades, the leaf-area band that defines each, and the photographs in each grade

Grade	Leaf area showing lesions, chlorosis or necrosis	Black Leaf Spot	Phyllosticta Leaf Spot	Total
mild	under 10%	52	57	109
moderate	10% to 40%	59	81	140
severe	over 40%	41	137	178

Grade	Leaf area showing lesions, chlorosis or necrosis	Black Leaf Spot	Phyllosticta Leaf Spot	Total
Total		152	275	427

Two tools protected the labels. The label-table generator carries every label already entered into a regenerated table and keeps a dated backup of the previous file, so rerunning it can never erase work; and a progress report lists every row still awaiting a grade, so no photograph could be skipped. No second person graded the photographs, so agreement between raters is not reported (Section 7.5). [Describe how you estimated the affected proportion of each leaf.]

Appendix C Treatment knowledge base

Table C.1 summarises the advice held for every outcome the cascade can return. Each chemical option is listed with its FRAC group so that a grower can rotate between groups. At the time of writing no rate has been verified, so the application shows a referral to the product label and an extension officer in place of every dose (Figure 6.8).

Table C.1: Advice held for each outcome, with chemical options and expert referral

Outcome and grade	Summary shown to the grower	Chemical options	Refer to expert
Black Leaf Spot, mild	A few small lesions. Cultural correction is usually enough; no spraying needed yet.	none recommended	no
Black Leaf Spot, moderate	Established infection spreading across the leaf. Sanitation plus a protectant fungicide.	Mancozeb, FRAC group M03 (multi-site protectant); Thiophanate-methyl, FRAC group 1 (MBC); rates unverified	no
Black Leaf Spot, severe	Over 40% of leaf area lost. The plant is at risk; treat aggressively and consider whether it is worth saving.	Boscalid + Pyraclostrobin, FRAC group 7 + 11; Thiophanate-methyl, FRAC group 1 (MBC); rates unverified	yes
Phyllosticta Leaf Spot, mild	Scattered small flecks. Correct the watering and airflow before reaching for chemicals.	none recommended	no
Phyllosticta Leaf Spot, moderate	Lesions coalescing and spreading. Sanitation plus a protectant or systemic fungicide.	Mancozeb, FRAC group M03 (multi-site protectant);	no

Outcome and grade	Summary shown to the grower	Chemical options	Refer to expert
		Chlorothalonil, FRAC group M05 (multi-site protectant); rates unverified	
Phyllosticta Leaf Spot, severe	Extensive leaf loss. Systemic treatment, and reassess whether the plant is worth keeping.	Boscalid + Pyraclostrobin, FRAC group 7 + 11; Thiophanate-methyl, FRAC group 1 (MBC); Azoxystrobin, FRAC group 11 (QoI); rates unverified	yes
Healthy	No disease detected. Advice here is preventive maintenance.	none	no
Unidentified condition	The plant does not appear healthy, but the system is not confident enough to name the condition. Do not treat on a guess.	none	yes
Refused photograph	This does not look like a Vanda orchid, so no diagnosis was attempted.	none	no

Conditions outside the classifier's scope are kept in the knowledge base so that the distinctions that matter for treatment are never lost (Table C.2).

Table C.2: Out-of-scope conditions retained in the knowledge base

Condition	Pathogen type	Recorded guidance
Phytophthora black rot	oomycete	An oomycete, not a true fungus: standard fungicides do not work; metalaxyl-class products and cinnamon paste on cut surfaces are recorded
Anthracnose	fungus	Out of scope for the classifier; needs its own treatment entry before the classifier may predict it
Sooty mould	fungus	A symptom of sap-sucking insects; the insects are the problem to control

Every chemical recommendation is shown with the following safety notes:

- Read the product label before mixing. The label is the legal instruction.
- Wear gloves, mask and eye protection when mixing or spraying.
- Observe the pre-harvest and re-entry interval printed on the label.
- Do not spray in direct midday sun or when temperature is above 30 C -- Vanda leaves scorch.
- Test any new product on two or three plants and wait seven days before treating a whole collection.

- Rotate between fungicide groups with different FRAC codes. Using one active ingredient repeatedly selects for resistant strains and the product stops working.

Appendix D Input check calibration record

Table D.1 lists the feature distance and the decision for every non-orchid photograph, measured by the deployed cascade on 10 October 2026 with the threshold of 0.55. Table D.2 summarises the distances of each set, both at calibration and in that run.

Table D.1: Feature distance and decision for each non-orchid photograph

Photograph	Content	Distance	Decision
grass2.jpg	grass	0.469	accepted, diagnosed healthy
tree3.jpg	tree	0.515	accepted, diagnosed healthy
tree1.jpg	tree	0.563	refused
grass1.jpg	grass	0.584	refused
IMG-20240501-WA0015.jpg	messaging-application image	0.589	refused
dog1.jpg	dog	0.601	refused
tree2.jpg	tree	0.602	refused
IMG-20240501-WA0257.jpg	messaging-application image	0.610	refused
table3.jpg	table	0.615	refused
table2.jpg	table	0.618	refused
dog2.jpg	dog	0.648	refused
hand2.jpg	hand	0.665	refused
dog3.jpg	dog	0.673	refused
hand1.jpg	hand	0.677	refused
Screenshot (554).png	screenshot	0.696	refused
IMG-20240501-WA0348.jpg	messaging-application image	0.718	refused
table1.jpg	table	0.725	refused

Table D.2: Feature distances of each set, with the number refused at the 0.55 threshold

Set	Photographs	Median distance	Largest distance	Refused
Validation photographs, at calibration	67	0.296	0.410	0
Test photographs, at calibration	67	0.300	0.451	0
Test photographs, end-to-end run	67	0.300	0.453	0
Non-orchid photographs, end-to-end run	17	0.615	0.725	15

The eleven orchid disease photographs taken from the web lay between 0.313 and 0.547 at calibration, and all were accepted. The difference between the two test-set

measurements of the largest distance, 0.451 and 0.453, comes from the different image-decoding libraries used by the calibration script and the deployed cascade.

Appendix E Interface and data contract specification

All endpoints sit under /api/v1/disease. Table E.1 gives the request and response of each, and Table E.2 explains the fields of a diagnosis result.

Table E.1: Endpoints of the disease router, with their request and response contracts

Endpoint	Request	Response
POST /detect	multipart form: image (an image file of up to 12 MB); optional query parameter threshold between 0 and 1	status, filename and result (Table E.2); HTTP 400 for an unusable file; HTTP 503 when a model is missing
GET /treatments/{disease}	path: disease name; optional query parameter severity (mild, moderate or severe)	status and treatment; an unknown name returns the unidentified advice
GET /classes	none	status, diseases (name, display name and grades of each) and a note on what is predicted
GET /status	none	status (success or degraded) and detail: which model files are present, whether each model is ready, and the threshold in force
POST /contribute	multipart form: image, disease, plant_part (leaf, stem or flower), severity, verified, verified_by, notes	status and result: saved, image_id, disease, verified, count_verified, minimum_required, needed, ready_for_training; HTTP 422 for an incomplete or inconsistent form
GET /pending-counts	none	status, minimum_required, and for each disease: total, verified, needed and ready_for_training

Table E.2: Fields of a diagnosis result

Field	Type	Meaning
disease	string	black_leaf_spot, phyllosticta_leaf_spot, healthy, unidentified or invalid_image
raw_prediction	string or null	the most probable class before the threshold is applied; withheld for a refused photograph
confidence	number or null	probability of the most probable class
confident	boolean	whether the confidence reaches the threshold
threshold	number	the confidence threshold in force, 0.70 by default
plant_part	string	leaf, because both trained diseases are leaf diseases
probabilities	object	probability of each trained class
severity	string or null	mild, moderate or severe; present only for a confident disease
severity_confidence	number or null	probability of the predicted grade
severity_probabilities	object or null	probability of each grade
severity_note	string	present when the grade's confidence is below 0.55
valid_orchid_image	boolean	false when the input check refused the photograph
validation_distance	number	feature distance to the five most similar training

Field	Type	Meaning
		photographs
validation_threshold	number	the input check threshold, 0.55
explanation	string	a plain-language reason for the outcome
treatment	object	advice from the knowledge base: summary, immediate actions, cultural control, chemical control, monitoring, expert referral and safety notes

Appendix F Verification evidence and research artefact inventory

Verification of this component rests on three kinds of evidence: the scripted checks listed in Table 5.2, the held-out evaluations reported in Chapter 6, and the end-to-end. There is no automated unit-test suite for the component at the time of submission. The endpoint behaviors in Table 6.7 were exercised against a running service and are reported as observed behavior rather than as a pass rate. Adding them to an automated suite is listed as future work.

What does exist is a record of every measurement quoted in this thesis. Table F.1 lists each artefact, its contents and the result it supports, so that no number in Chapter 6 rests on a figure that cannot be traced to a file. Paths are relative to ml-models/disease_detection in the project repository.

Table F.1: Research artefacts, their contents and the results they support

Artefact	Contents	Supports
data/raw/PROVENANCE.md	renames applied to the raw photographs and why	Section 3.3.1
data/split/split_manifest.csv	the split of every original photograph	Table 3.1
data/leakage_report.txt	the three leakage checks	Figure 3.2; Table 5.2
data/severity_labels.csv	class, split and grade of every original photograph	Tables 3.1 and 6.5; Appendix B
models/training_metadata.json, training_history.json	configuration, class weights and per-epoch history	Table 3.2; Figure 6.2
models/metrics_test.json, metrics_validation.json, evaluation_report_test.txt	per-class metrics, confusion matrices, threshold sweeps and misclassified photographs	Tables 6.1 and 6.3; Figures 6.1 and 6.4; Appendix A
models/model_comparison.json	the backbone comparison	Table 6.2; Figure 6.3
models/severity_metrics_test.json, severity_training_metadata.json	severity metrics and configuration	Table 6.4; Figure 6.5
models/feature_bank.npz, feature_bank_metadata.json	reference vectors, threshold and calibration results	Section 3.3.6; Figure 6.6; Appendix D
src/treatment_kb.json	treatment advice and dose verification flags	Section 4.4.3; Appendix C
End-to-end run record, 10 October 2026	outcome, confidence, grade and distance of 84 photographs	Table 6.6; Appendix D
Endpoint test record, 10 October 2026	the 14 requests and their responses	Table 6.7
Application screenshots, 11 October 2026	screens captured from the running system	Figures 4.3, 6.7 and 6.8

Appendix G Academic integrity and AI use disclosure

This appendix discloses the use of generative artificial intelligence and other automated tools in developing the component and preparing this thesis, as the module policy requires. The candidate declaration applies in full: whatever tools were used, the author remains responsible for the accuracy, originality and conclusions of everything reported here. Table G.1 records tool use by activity, with the verification carried out in each case.

Table G.1: Use of generative AI tools by activity, with the verification performed

Activity	Tool and purpose	Verification performed
Photograph collection and labelling	No AI tool used. The photographs were taken by the author, and every disease and severity label was assigned by the author	Not applicable
Data preparation, training, evaluation and verification scripts	A generative AI assistant used for debug the orientation, augmentation, leakage-check, training and evaluation scripts. The author ran the training on Google Colab. Model choices and thresholds rest on the measurements in Chapter 6: the assistant helped design the procedures, but no value was chosen without a measurement	The author ran every script; every reported number is read from the files in Appendix F
Backend service and mobile screens	Claude assisted in writing and debugging the router, service and screens	Code read and run by the author; endpoints tested (Table 6.7); screenshots taken from the running system
Figures and screenshots	Claude wrote the plotting scripts and the browser automation used for the screenshots	Each figure compared with the file it plots; no number was typed into a figure by hand
Literature and references	[State how the sources were found.] Claude formatted the references and checked several against their publishers' pages	[Confirm that you located and checked every reference]
Thesis text	Claude drafted and edited text under the author's direction	Every statement checked by the author against the code, the data and the evaluation outputs

The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this thesis.